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Ted Kaczynski sprinted through high school, not bothering with his junior year and making only passing efforts at social contact. Off to Harvard at age 16, Kaczynski was a loner during his college years. One of his roommates at Harvard said that he avoided people by quickly shuffling by them and slamming the door behind him. After obtaining his Ph.D. in mathematics at the University of Michigan, Kaczynski became a professor at the University of California at Berkeley. His colleagues there remember him as hiding from social interaction—no friends, no allies, no networking.

After several years at Berkeley, Kaczynski resigned and moved to a rural area of Montana, where he lived as a hermit in a crude shack for 25 years. Town residents described him as a bearded eccentric. Kaczynski traced his own difficulties to growing up as a genius in a kid's body and sticking out like a sore thumb in his surroundings as a child. In 1996, he was arrested and charged as the notorious Unabomber, America's most wanted killer. Over the course of 17 years, Kaczynski had sent 16 mail bombs that left 23 people wounded or maimed and 3 people dead. In 1998, he pleaded guilty to the offenses and was sentenced to life in prison.

A decade before Kaczynski mailed his first bomb, Alice Walker spent her days battling racism in Mississippi. She had recently won her first writing fellowship, but rather than use the money to



Ted Kaczynski, the convicted Unabomber, traced his difficulties to growing up as a genius in a kid's body and not fitting in when he was a child.

(Top) Seanna O'Sullivan; (bottom) WBBM-TV/AFP/Getty Images

follow her dream of moving to Senegal, Africa, she put herself into the heart and heat of the civil rights movement. Walker had grown up knowing the brutal effects of poverty and racism. Born in 1944, she was the eighth child of Georgia sharecroppers who earned \$300 a year.

When Walker was 8, her brother accidentally shot her in the left eye with a BB gun. Since her parents had no car, it took them a week to get her to a hospital. By the time she received medical care, she was blind in that eye, and it had developed a disfiguring layer of scar tissue. Despite the counts against her, Walker overcame pain and anger and went on to win a Pulitzer

Prize for her book *The Color Purple*. She became not only a novelist but also an essayist, a poet, a short-story writer, and a social activist.

What leads one individual, so full of promise, to commit brutal acts of violence and another to turn poverty and trauma into a rich literary harvest? If you have ever wondered why people turn out the way they do, you have asked yourself the central question we will explore in this release.

Essentials of Life-Span Development is a window into the journey of human development—your own and that of every other member of the human species. Every life is distinct, a new biography in the world. Examining the shape of life-span development helps us to understand it better. In this chapter, we explore what it means to take a life-span perspective on development, examine the nature of development, and outline how science helps us to understand it. ■



Alice Walker won the Pulitzer Prize for her book *The Color Purple*. Like the characters in her book, Walker overcame pain and anger to triumph and celebrate the human spirit.

(Top) Monica Morgan/WireImage/Getty Images; (bottom) Courtesy of Alice Walker

The Life-Span Perspective

Each of us develops partly like all other individuals, partly like some other individuals, and partly like no other individual. Most of the time we notice the qualities in an individual that make that person unique. But as humans, we have all traveled some common paths. Each of us—Leonardo da Vinci, Joan of Arc, George Washington, Martin Luther King, Jr., and you—walked at about 1 year, engaged in fantasy play as a young child, and became more independent as a youth. Each of us, if we live long enough, will experience hearing problems and the death of family members and friends. This is the general course of our **development**, the pattern of movement or change that begins at conception and continues through the human life span.

In this section we explore what is meant by the concept of development and why the study of life-span development is important. We outline the main characteristics of the life-span perspective and discuss various influences on development. In addition, we examine some contemporary concerns related to life-span development.

The Importance of Studying Life-Span Development

How might you benefit from studying life-span development? Perhaps you are, or will be, a parent or teacher. If so, responsibility for children is, or will be, a part of your everyday life. The more you learn about them, the better you can raise them or teach them. Perhaps you hope to gain some insight

development The pattern of movement or change that starts at conception and continues through the life span.

about your own history—as an infant, a child, an adolescent, or a young adult. Perhaps you want to know more about what your life will be like as you grow through the adult years—as a middle-aged adult, or as an adult in old age, for example. Or perhaps you just stumbled across this course, thinking that it sounded intriguing. Whatever your reasons, you will discover that the study of life-span development addresses some provocative questions about who we are, how we came to be this way, and where our future will take us.

In our exploration of development, we will examine the life span from the point of conception until the time when life (at least, life as we know it) ends. You will see yourself as an infant, as a child, and as an adolescent, and you will learn about how those years influenced the kind of individual you are today. And you will see yourself as a young adult, as a middle-aged adult, and as an adult in old age, and you may be motivated to consider how your experiences will affect your development through the remainder of your adult years.

Characteristics of the Life-Span Perspective

Growth and development are dramatic during the first two decades of life, but development is not something that happens only to children and adolescents (Yang & others, 2023). The traditional approach to the study of development emphasizes extensive change from birth to adolescence (especially during infancy), little or no change in adulthood, and decline in old age. Yet a great deal of change does occur in the decades after adolescence. The life-span approach emphasizes developmental change throughout adulthood as well as during childhood (Sherazi & others, 2023).

Life Expectancy

Recent increases in human life expectancy have contributed to greater interest in the life-span approach to development. The upper boundary of the human life span (based on the oldest age documented) is 122 years. The maximum life span of humans has not changed since the beginning of recorded history. What has changed is life expectancy, the average number of years that a person born in a particular year can expect to live. In the twentieth century alone, life expectancy increased by 30 years, thanks to improvements in sanitation, nutrition, and medicine (see Figure 1). For individuals born in 2021 in the United States, their life expectancy was 76.1 years of age (National Center for Health Statistics, 2022). Today, for most individuals in developed countries, childhood and adolescence represent only about one-fourth of their lives.

For the first time in U.S. history, in 2019 there were more people over 60 years of age than under 18 years of age. In less than a century, more years were added to human life expectancy than in all of the prior millennia.

However, because of the staggering number of deaths due to the COVID-19 pandemic, U.S. life expectancy dropped by 1.8 years from 2019 to 2020, going from 78.8 years in 2019 to 77.0 years in 2020 (National Center for Health Statistics, 2022). A disproportionate number of deaths due to COVID-19 occurred among Black Americans and Latinos. For Black Americans, life expectancy dropped by 3.2 years, going from 74.7 years in 2019 to 71.5 years in 2020, and for Latinos, it declined by 3.9 years, going from 81.8 years in 2019 to 77.9 years in 2020. COVID-19 was responsible for 90 percent of the declining life expectancy among Latinos. The overall U.S. life expectancy declined further from 2020 to 2021. That decline—from 77.0 years in 2020 to 76.1 years in 2021—took U.S. life expectancy at birth to its lowest level since 1996. The 0.9 year drop in life expectancy in 2021, along with a 1.8 year drop in 2020, was the biggest two-year decline in life expectancy since 1921–1923 (National Center for Health Statistics, 2022).

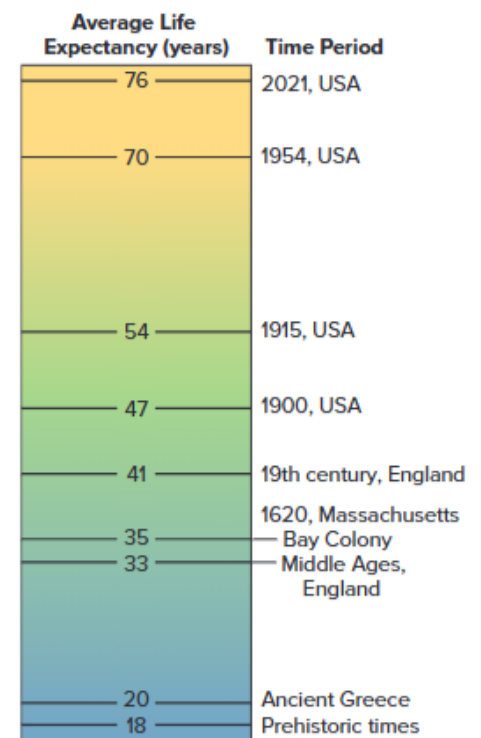


Figure 1 Human Life Expectancy at Birth from Prehistoric Time to Contemporary Times



How has the COVID-19 pandemic affected older adults?
triloks/Getty Images

After surfacing near the end of 2019 and spreading rapidly in 2020 and 2021, the COVID-19 pandemic disrupted daily life and created high levels of stress for large portions of the world's population. The disease affected older adults more than any other age group, with people 65 years of age and older contracting more severe and deadly versions of COVID-19 than younger individuals. The pandemic especially attacked older adults living in nursing homes and assisted living facilities, where they were highly vulnerable to transmitting the disease to each other.

Another hazard of COVID-19 for older adults has been the increased loneliness and anxiety caused by isolation from family and friends during periods of quarantine (Ribeiro-Goncalves & others, 2023).

Several years ago, Laura Carstensen (2015, 2016, 2019) described the challenges and opportunities created by dramatic increases in life expectancy in recent decades. In her view, the remarkable increase in the number of people living to old age has taken place so quickly that science, technology, and behavioral changes have not kept pace. She proposes that the challenge is to transform a world constructed mainly for young people into a world that is more compatible and supportive for the increasing number of people living to 100 and older.

In further commentary, Carstensen (2015, p. 70) remarked that making this transformation would be no small feat:

... parks, transportation systems, staircases, and even hospitals presume that the users have both strength and stamina; suburbs across the country are built for two parents and their young children, not single people, multiple generations or elderly people who are not able to drive. Our education system serves the needs of young children and young adults and offers little more than recreation for experienced people.

Indeed, the very conception of work as a full-time endeavor ending in the early sixties is ill suited for long lives (Carstensen, 2023). Arguably the most troubling aspect of our current perspective on aging is that we fret about ways that older people lack the qualities of younger people rather than exploit a growing new resource right before our eyes: citizens who have deep expertise, emotional balance, and the motivation to make a difference (Carstensen, 2023).

Certainly, recent progress has been made in improving the lives of older adults. In our discussion of late adulthood, you will read about researchers who are exploring ways to modify the activity of genes related to aging, methods for improving brain functioning in older people, medical discoveries for slowing or even reversing the effects of various chronic diseases, and ways to prepare for a better quality of life when we get old, including strategies for staying cognitively sharp, maintaining our physical fitness, and becoming more satisfied with our lives as older adults. But much more remains to be accomplished (Caplan, 2023).

Dimensions of the Life-Span Perspective

The belief that development occurs throughout life is central to the life-span perspective on human development, but this perspective has other characteristics as well. According to life-span development expert Paul Baltes (1939–2006), the **life-span perspective** views development as lifelong, multidimensional, multidirectional, plastic, multidisciplinary, and contextual, and as a process that involves growth, maintenance, and regulation of loss (Baltes, 1987, 2003; Baltes, Lindenberger, & Staudinger, 2006). In this view, it is important to understand that development is constructed through biological, sociocultural, and individual factors working together (Baltes, Reuter-Lorenz, & Rösler, 2006). Let's look at each of these characteristics.

Development Is Lifelong In the life-span perspective, early adulthood is not the endpoint of development; rather, no age period dominates development. Researchers increasingly study the experiences and psychological orientations of adults at different points in their lives. Later in this chapter we describe the age periods of development and their characteristics.

life-span perspective The perspective that development is lifelong, multidimensional, multidirectional, plastic, multidisciplinary, and contextual; that it involves growth, maintenance, and regulation; and that it is constructed through biological, sociocultural, and individual factors working together.

Development Is Multidimensional Development consists of biological, cognitive, and socioemotional dimensions. Even within each of those dimensions, there are many components (Kenderla & Kibbe, 2023). The cognitive dimension, for example, includes attention, memory, abstract thinking, speed of processing information, wisdom, and social intelligence. At every age, changes occur in every dimension. Changes in one dimension also affect development in the other dimensions.

To get an idea of how interactions occur, consider the development of Ted Kaczynski, the so-called Unabomber discussed at the beginning of the chapter. When he was 6 months old, he was hospitalized with a severe allergic reaction, and his parents were rarely allowed to visit him. According to his mother, the previously happy baby was never the same after his hospital stay. He became withdrawn and unresponsive. As Ted grew up, he had periodic “shutdowns” accompanied by rage. In his mother’s view, events that occurred during her son’s infancy warped the development of his mind and emotions.

Development Is Multidirectional Throughout life, some dimensions or components of a dimension expand and others shrink. For example, when one language (such as English) is acquired early in development, the capacity for acquiring second and third languages (such as Spanish and Chinese) decreases later in development, especially after early childhood (Levitt, 1989). During adolescence, as individuals establish romantic relationships, their relationships with friends might decrease. During late adulthood, older adults might become wiser by being able to call on experience to guide their intellectual decision making (Gluck & Westrate, 2022), but they perform more poorly on tasks that require speed in processing information (Anderson & others, 2022).

Development Is Plastic Even at 10 years old, Ted Kaczynski was extraordinarily shy. Was he destined to remain forever uncomfortable with people? Developmentalists debate how much plasticity people have in various dimensions at different points in their development (Flores & others, 2023). Plasticity means the capacity for change. For example, can you still improve your intellectual skills when you are in your seventies or eighties? Or might these intellectual skills be fixed by the time you are in your thirties so that further improvement is impossible? Researchers have found that the cognitive skills of older adults can be improved through cognitive or physical training (Belleville & others, 2023). However, possibly we possess less capacity for change when we become older. The exploration of plasticity and its constraints is a key element on the contemporary agenda for developmental research.

Developmental Science Is Multidisciplinary Psychologists, sociologists, anthropologists, neuroscientists, and medical researchers all share an interest in unlocking the mysteries of development through the life span. How do your heredity and health limit your intelligence? Do intelligence and social relationships change with age in the same way around the world? How do families and schools influence intellectual development? These are examples of research questions that cut across disciplines.

Development Is Contextual All development occurs within a **context**, or setting. Contexts include families, schools, peer groups, churches, cities, neighborhoods, university laboratories, countries, and so on. Each of these settings is influenced by historical, economic, social, and cultural factors.

Contexts, like individuals, change (Clauss-Ehlers & others, 2023). Thus, individuals are changing beings in a changing world. As a result of these changes, contexts exert three types of influences (Baltes, 2003): (1) normative age-graded influences, (2) normative history-graded influences, and (3) nonnormative or highly individualized life events. Each of these types of influences can have a biological or environmental impact on development.

Normative age-graded influences are similar for individuals in a particular age group. These influences include **biological processes** such as puberty and menopause. They also include sociocultural, environmental

context The setting in which development occurs, which is influenced by historical, economic, social, and cultural factors.

normative age-graded influences Biological and environmental influences that are similar for individuals in a particular age group.

biological processes Changes in an individual’s physical nature.



Nonnormative life events, such as the Russia-Ukraine war (see above) in Ukraine in 2022 and Hurricane Maria in Puerto Rico in 2017, are unusual circumstances that can have a major influence on a children's and adults' development.
Mykhaylo Palinchak/SOPA Images/
LightRocket/Getty Images

processes such as beginning formal education (usually at about age 6 in most cultures) and retirement (which takes place during the fifties and sixties in most cultures).

Normative history-graded influences are common to people of a particular generation because of historical circumstances (Elder & Cox, 2020). For example, in their youth American baby boomers shared the experience of the Cuban missile crisis, the assassination of John F. Kennedy, and the Beatles invasion. Other examples of normative history-graded influences include economic, political, and social upheavals such as the Great Depression in the 1930s, World War II in the 1940s, the civil rights and women's rights movements of the 1960s and 1970s, the terrorist attacks of 9/11/2001, the integration of computers and cell phones into everyday life during the 1990s, and time spent on social media in the twenty-first century. Long-term changes in the genetic and cultural makeup of a population (due to immigration or changes in fertility rates) are also part of normative historical change.

Nonnormative life events are unusual occurrences that have a major impact on the individual's life. These events do not happen to all people, and when they do occur they can influence people in different ways (Ponnusamy, 2022). Examples include the death of a parent when a child is young, pregnancy in early adolescence, a fire that destroys a home, a war, winning the lottery, or getting an unexpected career opportunity.

Development Involves Growth, Maintenance, and Regulation of Loss Baltes and his colleagues (2006) assert that the mastery of life often involves conflicts and competition among three goals of human development: growth, maintenance, and regulation of loss. As individuals age into middle and late adulthood, the quest to maintain their capacities and to regulate loss takes center stage away from growth. Thus, a 75-year-old man might aim not to improve his memory or his golf swing but to maintain his independence and to continue playing golf. In other chapters, we will discuss these ideas about maintenance and regulation of loss in greater depth.

Development Is a Co-Construction of Biology, Culture, and the Individual Development comes from biological, cultural, and individual factors influencing each other. For example, the brain shapes culture, but it is also shaped by culture and the experiences that individuals have or pursue (Koyama & others, 2023). In terms of individual factors, we can go beyond what our genetic inheritance and environment give us. We can create a unique developmental path by actively choosing from the environment the things that optimize our lives (Rathunde & Csikszentmihalyi, 2006).

How Would You...?

As a **social worker**, how would you explain the importance of asking for information about **nonnormative life events** when working with a new client?



normative history-graded influences Biological and environmental influences that are associated with history. These influences are common to people of a particular generation.

nonnormative life events Unusual occurrences that have a major impact on a person's life. The occurrence, pattern, and sequence of these events are not applicable to many individuals.

Contemporary Concerns in Life-Span Development

Pick up a magazine or visit a news site and you might see headlines like these: "Screen Time Linked to Children's Obesity," "Political Leanings May Be Written in the Genes," "Religious Group Protests Transgender Bathrooms," "FDA Warns About Side Effects of ADHD Drug," "COVID-19 Deaths Higher in African American Patients," "Test May Predict Alzheimer Disease." Researchers using the life-span perspective explore these and many other topics of contemporary concern. The roles that health and well-being, parenting, education, sociocultural contexts, and

Careers in life-span development

Gustavo Medrano, Clinical Psychologist

Gustavo Medrano specializes in helping children, adolescents, and adults of all ages improve their lives when they have problems involving depression, anxiety, emotion regulation, chronic health conditions, and life transitions. He works individually with clients and provides therapy for couples and families. As a native Spanish speaker, he also provides bicultural and bilingual therapy for clients.

Dr. Medrano is a faculty member in the Department of Psychiatry at the University of Illinois–Chicago. He obtained his undergraduate degree in psychology at Northwestern University and then became a high school teacher through Teach for America, a program where participants spend at least two years teaching in a high-poverty area. He received his master's and doctoral degrees in clinical psychology at the University of Wisconsin–Milwaukee. As a faculty member, in addition to doing clinical therapy with clients, he has conducted research focusing on how family experiences, especially parenting, influence children's and adolescents' ability to cope with chronic pain and other challenges.



Gustavo Medrano is a clinical psychologist who often works with Spanish-speaking clients.
Avis Mandel Pictures

technology play in life-span development, as well as how social policy is related to these issues, are a particular focus of this release.

Health and Well-Being

Health professionals today recognize the powerful influences of lifestyles and psychological states on health and well-being (Hahn & others, 2023). Clinical psychologists are among the health professionals who help people improve their well-being. To read about one clinical psychologist who helps adolescents and adults improve their developmental outcomes, see the *Careers in Life-Span Development* profile.

Parenting and Education

Can two gay men be successful parents? Do children suffer if their parents get divorced? Are U.S. schools failing to teach children how to read and write and calculate adequately? Questions like these reflect pressures on the contemporary family and the challenges faced by U.S. schools (Sadker & others, 2023). In later chapters, we analyze child care, the effects of divorce, parenting styles, intergenerational relationships, early childhood education, relationships between childhood poverty and education, children with disabilities, bilingual education, new educational efforts to improve lifelong learning, and many other issues related to parenting and education (Peregoy & Boyle, 2023; Trawick-Smith, 2023).

Sociocultural Contexts and Diversity

Health, parenting, and education—like development itself—are shaped by their sociocultural context (Telljohann & others, 2023). To analyze this context, four concepts are especially useful: culture, ethnicity, socioeconomic status, and gender.



Robert Maust/Photo Agora



Asian American and Latino children are the fastest-growing immigrant groups in the United States. How diverse are the students in your life-span development class? Are their experiences in growing up likely to have been similar to or different from yours?
Zuma Press Inc./Alamy Stock Photo



How Would You...?

As a health-care professional, what examples might you use to show that understanding culture, ethnicity, SES, and gender is essential in educating future health-care professionals?

culture The behavior patterns, beliefs, and all other products of a group that are passed on from generation to generation.

cross-cultural studies Comparisons of one culture with one or more other cultures. These provide information about the degree to which children's development is similar, or universal, across cultures, and the degree to which it is culture-specific.

ethnicity A range of characteristics rooted in cultural heritage, including nationality, race, religion, and language.

Culture encompasses the behavior patterns, beliefs, and all other products of a particular group of people that are passed on from generation to generation. Culture results from the interaction of people over many years (Halder & Squires, 2023). A cultural group can be as large as the United States or as small as an isolated Appalachian town. Whatever its size, the group's culture influences the behavior of its members.

Cross-cultural studies compare aspects of two or more cultures. The comparison provides information about the degree to which development is similar, or universal, across cultures, or is instead culture-specific (Chentsova & others, 2023). For example, in a study of 26 countries, individuals in Chile had the highest life satisfaction, those in Bulgaria and Spain the lowest (Jang & others, 2017).

Ethnicity (the word *ethnic* comes from the Greek word for "nation") is rooted in cultural heritage, nationality, race, religion, and language. Black Americans, Latinos, Asian Americans, Native Americans, European Americans, and Arab Americans are examples of broad ethnic groups in the United States. Diversity exists within each ethnic group (Garrod & Kilenny, 2023). In 2020, 50 percent of children 17 years and younger were non-Latino White; by 2050, this figure is projected to decrease to 39 percent (ChildStats.gov, 2022). In 2020 in the United States, 26 percent of children were Latino, but in 2050 that figure is projected to increase to 31 percent. Asian Americans are expected to be the fastest-growing ethnic group of children percentage-wise: In 2020, 5 percent were Asian American, and that figure is expected to grow to 7 percent in 2050. No change in the percentage of Black American children (14 percent) is anticipated from 2020 to 2050. A special concern is the discrimination and prejudice experienced by ethnic minority children (Gomez, 2023; Harris & Carpenter, 2023).

This changing demographic tapestry promises not only the richness that diversity brings but also the difficult challenges of extending the American dream to all individuals (Steele & Newton, 2023). In 2020, the deadly effects of racial discrimination took center stage in the United States as not only Black Americans but members of other groups protested the killing of Black Americans such as George Floyd by non-Latino White police officers.

Over historical time, the preferred terms for referring to ethnic and racial groups have changed. Research studies, professional organizations, the media, and individuals themselves have modified the way they refer to ethnic and racial groups. For example, in 2020, members of the Latino Caucus of the Society for Research on Child Development, a leading professional organization, voted to change the name to the Latinx Caucus, in part because Latinx was considered more inclusive of LGBTQ, gender diverse, and non binary members. In 2021, around the same time, the League of United Latin American Citizens, a civil rights organization, dropped the term *Latinx* from their official communication. The organization's president, Domingo Garcia, explained that the term was "very unliked" by most Latinos and was perceived as being primarily used in academic circles. Results of a Pew Research Center poll reflect this position. The poll found that only 23 percent of adults who identify themselves as Latino or Hispanic had even heard of the term Latinx, and of those who had heard the term, only 3 percent used it themselves (Noe-Bustamante, Mora, & Lopez, 2020).

Preferred terminology for identifying Black Americans has changed over time, too. The 2020 U.S. Census Bureau question includes the option "Black or African American" (U.S. Census Bureau, 2021). The shift in terminology from African American to Black American is intended to be



Doly Akter is from Dhaka, Bangladesh, where sewers overflow, garbage rots in the streets, and children are undernourished. Nearly two-thirds of the women in Bangladesh marry before they are 18. Doly organized a club supported by UNICEF in which girls go door-to-door to monitor the hygiene habits of households in their neighborhood, which has led to improved hygiene and health in the families. Also, her group has managed to stop several child marriages by meeting with parents and convincing them that it is not in their daughter's best interests. They emphasize the importance of staying in school and how this will improve their daughter's future. Doly says that the girls in her UNICEF group are far more aware of their rights than their mothers ever were (UNICEF, 2007).

Naser Siddique/UNICEF Bangladesh

more inclusive because many people, such as immigrants from the Caribbean, identify themselves as Black but not as African American. In this release, we generally use the terms Latino or Latina, Black American, and non-Latino White for the corresponding U.S. demographic groups. However, we recognize that these and other terms will not represent every individual's or group's point of view.

Socioeconomic status (SES) refers to a person's position within society based on occupational, educational, and economic characteristics. Socioeconomic status implies certain inequalities (Niemeyer & Knaevelsrud, 2023). Differences in the ability to control resources and to participate in society's rewards produce unequal opportunities.

Gender refers to a mosaic of categories related to femininity and masculinity based on social and cultural norms. Few aspects of our development are more central to our identity and social relationships than gender (Etaugh & Bridges, 2023; Matlin & Foushee, 2023). We discuss sociocultural contexts and diversity in each chapter.

Transgender is a broad term that refers to individuals who adopt a gender identity that differs from the one assigned to them at birth (Davis, 2023). A transgender identity of being born male but identifying with being female is much more common than the reverse (Zucker, Lawrence, & Kreukels, 2016). We will have much more to say about gender and transgender later in the text.

The conditions in which many of the world's women live are a serious concern (UNICEF, 2023). Inadequate educational opportunities, violence, and lack of political access are just some of the problems faced by many women.

In this section, you have read about important aspects of individuals such as their cultural and socioeconomic backgrounds, ethnicity, and gender. As people go through the life span, these categories intersect and can create systems of power and privilege as well as oppression and discrimination. For example, higher-SES non-Latino White males have experienced considerably greater privilege and less discrimination than Black American females. At various points in this text, we will further explore these category connections.

Social Policy

Social policy is a government's course of action designed to promote the welfare of its citizens. Values, economics, and politics all shape a nation's social policy. Out of concern that policy makers are doing too little to protect the well-being of children and older adults, life-span researchers are increasingly undertaking studies that they hope will lead to effective social policy (American Psychological Association, 2022).

Children who grow up in poverty represent a special concern. In 2021, 15.3 percent of U.S. children under 18 years of age were living in families with incomes below the poverty line, a decrease from 16.0 in 2020 (U.S. Census Bureau, 2022). This is a decrease from a peak of 23 percent in 1993 and also down from 19.7 percent in 2015. However, the U.S. figure of 15.3 percent of children living in poverty is much higher than the rates in other developed countries. For example, Canada has a child poverty rate of 9 percent and Sweden has a rate of 2 percent (Fontenot, Semega, & Kollar, 2018).

Compared with non-Latino White children, ethnic minority children are more likely to experience persistent poverty over many years and to live in isolated, poor neighborhoods where social supports are minimal and threats to positive development are abundant (Thomas & others, 2023).

socioeconomic status (SES) Refers to the conceptual grouping of people with similar occupational, educational, and economic characteristics.

gender A mosaic of characteristics related to femininity and masculinity based on social and cultural norms.

social policy A national government's course of action designed to promote the welfare of its citizens.

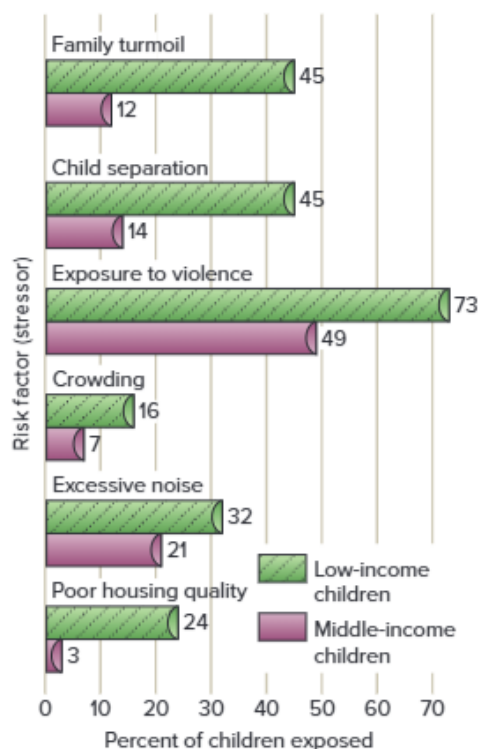


Figure 2 Exposure to Six Stressors Among Children in Low- and Middle-Income Families.

One study analyzed exposure to six stressors among children in low- and middle-income families (Evans & English, 2002). Low-income children were much more likely to face each of these stressors.

In 2021, Black American (25.9 percent, down from 36 percent in 2015) and Latino (22.4 percent, down from 30 percent in 2015) families with children had especially high rates of poverty (U.S. Census Bureau, 2022).

What are some of the negative experiences that children in poverty encounter? As indicated in Figure 2, one study found that a higher percentage of children in poor families than in middle-income families were exposed to family turmoil, separation from a parent, violence, crowding, excessive noise, and poor housing (Evans & English, 2002).

Developmental psychologists are seeking ways to help families living in poverty improve their well-being, and they have offered many suggestions for improving government policies. For example, there is increasing interest in developing two-generation educational interventions to improve the academic success of children living in poverty. For example, a recent large-scale effort to help children escape from poverty is the Ascend two-generation educational intervention being conducted by the Aspen Institute (Ascend, 2019, 2023). The intervention emphasizes education (promoting postsecondary education for mothers and improving the quality of children's early childhood education), economic support (housing, transportation, financial education, health insurance, and food assistance), and

social capital (peer support, including friends and neighbors; participation in community and faith-based organizations; school and work contacts). In a report, the Ascend program emphasized the importance of parents' education, economic stability, and overall health for their children's well-being (Ascend, 2019).

Some children triumph over poverty or other adversities. They show *resilience* (Sharp & Marais, 2022; Zhao & others, 2023). Think back to the chapter-opening story about Alice Walker. In spite of racism, poverty, her low socioeconomic status, and a disfiguring eye injury, she went on to become a successful author and champion for equality.

Are there certain characteristics that make children like Alice Walker resilient? Are there other characteristics that influence children like Ted Kaczynski, who despite his intelligence and education, became a killer? After analyzing research on this topic, Ann Masten and her colleagues (Masten, 2019, 2021; Masten & others, 2021) have concluded that a number of individual factors, such as good intellectual functioning, influence resiliency. In addition, family and extrafamilial contexts of

resilient individuals tend to share certain features. For example, resilient children are likely to have a close relationship to a caring parent figure and bonds to caring adults outside the family.

At the other end of the life span, protecting the well-being of older adults also creates policy issues (Sharp, 2023). Key concerns are escalating health care costs and insufficient access of older adults to adequate health care (McDuffie & Crowther, 2023).

Concerns about the well-being of older adults are heightened by two facts. First, the number of older adults in the United States is



What characterizes the Ascend program and how does it support families?
Tetra Images, LLC / Alamy Stock Photo

growing rapidly. Second, many of these older Americans are likely to need society's help.

Not only is the population of older adults growing in the United States, but also the world's population of people 60 years and older is projected to increase from 962 million in 2017 to 2.1 billion in 2050 (United Nations, 2017). The world's population 80 years of age and older is expected to triple or quadruple in this time frame. These significant increases in the world's older population have important implications for many sectors of society. When an increasing percentage of a country's population is older, governments must develop innovative policies and services that include improved housing, employment, health care, and transportation for older adults.



Ann Masten (*far right*) with a homeless mother and her child who are participating in her research on resilience. She and her colleagues have found that good parenting skills and good cognitive skills (especially attention and self-control) improve the likelihood that children in challenging circumstances will do better when they enter elementary school.

Dawn Vilella Photography

Technology

A final focus in our exploration of contemporary topics is the recent dramatic, almost overwhelming increase in the use of technology at all points in the life span (Hughes & Roblyer, 2023). From the introduction of television in the mid-1950s to today's widespread access to personal computers, the Internet, smartphones, and social media, as well as the ever-expanding use of robots in medicine and industry, it is obvious that our way of life has been forever changed by technological advances.

We will explore many technology topics in this release. Later in this chapter you will read about the emerging field of developmental robotics in our discussion of information processing as well as our coverage of different generations, including the extensive connection with technology of the millennial generation and Generation Z. At various points in the book, we explore topics such as whether babies should be allowed to watch television and videos, and especially how these activities might impair language development; how too much screen time takes away from children's exercise and increases their risks of obesity and cardiovascular disease; whether multitasking with electronic devices is helpful or harmful to academic success; as well as the extent to which older adults are adapting to the expanding role of technology in their daily lives.

How Would You...?

As a social worker, what scholarly information might you draw on to encourage social work majors in college to learn more about the needs of older adults?



The Nature of Development

In this section, we explore what is meant by developmental processes and periods, as well as variations in the way age is conceptualized. We examine some key developmental issues.

If you wanted to describe how and why Alice Walker or Ted Kaczynski developed during their lifetimes, how would you go about it? A chronicle of the events in any person's life can quickly become a confusing and tedious array of details. Two concepts help provide a framework for describing and understanding an individual's development: developmental processes and periods.

Biological, Cognitive, and Socioemotional Processes

At the beginning of this chapter, we defined development as the pattern of change that begins at conception and continues through the life span. The pattern is complex because it is the product of biological, cognitive, and socioemotional processes.

Biological Processes

Biological processes produce changes in an individual's physical nature. Genes inherited from parents, the development of the brain, height and weight gains, changes in motor skills, nutrition, exercise, the hormonal changes of puberty, and cardiovascular decline are all examples of biological processes that affect development.

Cognitive Processes

Cognitive processes refer to changes in an individual's thinking, intelligence, and language. Watching a colorful mobile swinging above the crib, putting together a two-word sentence, memorizing a poem, imagining what it would be like to be a movie star, and solving a crossword puzzle all involve cognitive processes.

Socioemotional Processes

Socioemotional processes involve changes in an individual's relationships with other people, in emotional regulation, and in personality. An infant's smile in response to a parent's touch, a toddler's aggressive attack on a playmate, a school-age child's development of assertiveness, an adolescent's joy at the senior prom, and the affection of an elderly couple all reflect the role of socioemotional processes in development.

Connecting Biological, Cognitive, and Socioemotional Processes

Biological, cognitive, and socioemotional processes are inextricably intertwined. Consider a baby smiling in response to a parent's touch. This response depends on biological processes (the physical nature of touch and responsiveness to it), cognitive processes (the ability to understand intentional acts), and socioemotional processes (the act of smiling often reflects a positive emotional feeling, and smiling helps to connect us in positive ways with other human beings). Nowhere is the connection across biological, cognitive, and socioemotional processes more obvious than in two rapidly emerging fields:

- *developmental cognitive neuroscience*, which explores links between cognitive processes, development, and the brain (Buzzell & others, 2023)
- *developmental social neuroscience*, which examines connections between socioemotional processes, development, and the brain (Bell, 2023)

In many instances, biological, cognitive, and socioemotional processes are bidirectional. For example, biological processes can influence cognitive processes and vice versa. For the most part, we will study the different processes of development (biological, cognitive, and socioemotional) in separate chapters, but the human being is an integrated individual with a mind and body that are interdependent. Thus, in many places throughout the book we will call attention to the connections between these processes.

biological processes Changes in an individual's physical nature.

cognitive processes Changes in an individual's thought, intelligence, and language.

socioemotional processes Changes in an individual's relationships with other people, emotions, and personality.

Periods of Development

The interplay of biological, cognitive, and socioemotional processes (see Figure 3) over time gives rise to the developmental periods of the human life span. A developmental period is a time frame in a person's life that is characterized by certain features. The most widely used

classification of developmental periods involves an eight-period sequence. For the purposes of organization and understanding, this book is structured according to these developmental periods.

The *prenatal period* is the time from conception to birth. It involves tremendous growth—from a single cell to a complete organism with a brain and behavioral capabilities—and takes place in approximately a nine-month period.

Infancy is the developmental period from birth to 18 or 24 months when humans are extremely dependent on adults. During this period, many psychological activities—language, symbolic thought, sensorimotor coordination, and social learning, for example—are just beginning.

Early childhood is the developmental period from the end of infancy to age 5 or 6. This period is sometimes called the “preschool years.” During this time, young children learn to become more self-sufficient and to care for themselves. They also develop school readiness skills, such as the ability to follow instructions and identify letters, and they spend many hours playing with peers. First grade typically marks the end of early childhood.

Middle and late childhood is the developmental period from about 6 to 11 years of age, approximately corresponding to the elementary school years. During this period, children master the fundamental skills of reading, writing, and arithmetic. They are formally exposed to the world outside the family and to the prevailing culture. Achievement becomes a more central theme of the child’s world, and self-control increases.

Adolescence encompasses the transition from childhood to early adulthood, entered at approximately 10 to 12 years of age and ending at 18 to 22 years of age. Adolescence begins with rapid physical changes—dramatic gains in height and weight, changes in body contour, and the development of sexual characteristics such as enlargement of the breasts, growth of pubic and facial hair, and deepening of the voice. At this point in development, the pursuit of independence and an identity are prominent themes. Thought is more logical, abstract, and idealistic. More time is spent outside the family.

Recently there has been increased interest in the transition between adolescence and adulthood, a transition that has been referred to as *emerging adulthood* (Arnett, 2016a, b). Emerging adulthood occurs approximately from 18 to 25 years of age and is a time of considerable exploration and experimentation, especially in the areas of identity, careers, and lifestyles (LeBlanc & Lyons, 2022).

Early adulthood is the developmental period that begins in the late teens or early twenties and lasts through the thirties. For young adults, this is a time for establishing personal and economic independence, becoming proficient in a career, and for many, selecting a mate, learning to live with that person in an intimate way, starting a family, and rearing children.

Middle adulthood is the developmental period from approximately 40 years of age to about 60. It is a time of expanding personal and social involvement and responsibility; of assisting the next generation in becoming competent, mature individuals; and of achieving and maintaining satisfaction in a career.

Late adulthood is the developmental period that begins in the sixties or seventies and lasts until death. It is a time of life review, retirement from the workforce, and adjustment to new social roles involving decreasing strength and health.

Late adulthood potentially lasts longer than any other period of development. Because the number of people in this age group has been increasing dramatically, lifespan developmentalists have been paying more attention to differences within late adulthood (Croize-Pourcelet & others, 2022). According to Paul Baltes and Jacqui Smith (2003), a major change takes place in older adults’ lives as they become the “oldest-old,” at about 85 years of age. The “young-old” (classified as 65 through 84 years of age in this analysis) have substantial potential for physical and cognitive

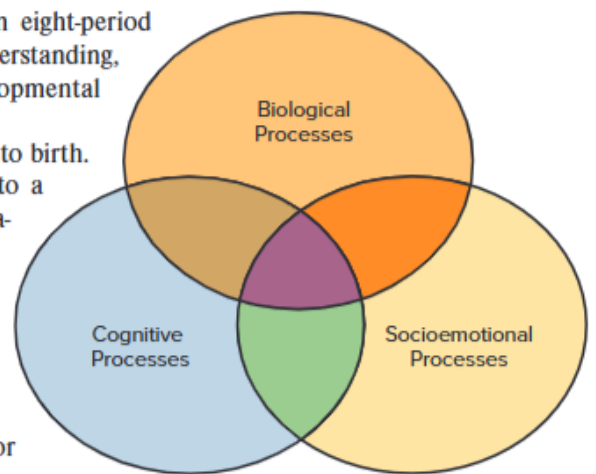


Figure 3 Processes Involved in Developmental Changes.

Biological, cognitive, and socioemotional processes interact as individuals develop.

fitness, retain much of their cognitive capacity, and can develop strategies to cope with the gains and losses of aging. In contrast, the oldest-old (85 and older) show considerable loss in cognitive skills, experience increased rates of chronic stress, and are more frail (Baltes & Smith, 2003). Nonetheless, considerable variation exists in how much of their capabilities the oldest-old retain (Wang & others, 2022).

Conceptions of Age

In our description of developmental periods, we attached an approximate age range to each period. But we also have noted that there are variations in the capabilities of individuals of the same age, and we have seen how age-related changes can be exaggerated. How important is age when we try to understand an individual?

According to some life-span experts, chronological age is not very relevant to understanding a person's psychological development (Chen & others, 2023). Chronological age is the number of years that have elapsed since birth. But time is a crude index of experience, and it does not cause development. Chronological age, moreover, is not the only way to measure age (Sherazi & others, 2023). Just as there are different domains of development, there are different ways of thinking about age.

Four Types of Age

Age has been conceptualized not just as chronological age but also as biological age, psychological age, and social age. *Biological age* is a person's age in terms of biological health. Determining biological age involves knowing the functional capacities of a person's vital organs. One person's vital capacities may be better or worse than those of others of comparable chronological age. The younger the person's biological age, the longer the person is expected to live, regardless of chronological age. One study involving 17-year survival rates of 20- to 93-year-old Korean adults found that death rates were higher among individuals whose biological age was greater than their chronological age (Yoo & others, 2017).

Psychological age is an individual's adaptive capacities compared with those of other individuals of the same chronological age. Thus, older adults who continue to learn, remain flexible, are motivated, think clearly, and have positive personality traits are engaging in more adaptive behaviors than their chronological age-mates who do not do these

things (Seifert & others, 2022). For example, a recent study found that older adults who were more conscientious and emotionally stable were less cognitively vulnerable (Duchek & others, 2020). And recent research revealed that psychological well-being, as indexed by having a higher purpose in life, positive affect, optimism, social support, and life satisfaction, predicted a lower mortality risk (Morozink Boylan, Tompkins, & Krueger, 2022).

Social age refers to connectedness with others and the social roles individuals adopt. Individuals who have better social relationships with others are happier and tend to live longer than individuals who are lonely (Singh & others, 2022).



How Would You...?

As a human development and family studies professional, what examples might you use to show that an understanding of the four ages is important in evaluating older adults?

(Left): Seventy-four-year-old Barbara Jordan participating in the long jump competition at a Senior Games in Maine; (right): a sedentary overweight middle-aged man. Even though Barbara Jordan's chronological age is older, might her biological age be younger than the middle-aged man's?

(Left) John Patriquin/Portland Press Herald/Getty Images; (right) Firehorse/E+/Getty Images



From a life-span perspective, an overall age profile of an individual involves not just chronological age but also biological age, psychological age, and social age. For example, a 70-year-old man (chronological age) might be in good physical health (biological age) but might be experiencing memory problems and having trouble coping with the demands placed on him by his wife's recent hospitalization (psychological age) and dealing with a lack of social support (social age).

Three Developmental Patterns of Aging

K. Warner Schaie (2016) recently described three developmental patterns that provide a portrait of how aging can involve individual variations:

- *Normal aging* characterizes most individuals, for whom psychological functioning often peaks in early middle age, remains relatively stable until the late fifties to early sixties, and then shows a modest decline through the early eighties. However, marked decline can occur as individuals near death.
- *Pathological aging* characterizes individuals who show greater than average decline as they age through the adult years. In early old age, they may have mild cognitive impairment, develop Alzheimer disease later on, or have a chronic disease that impairs their daily functioning.
- *Successful aging* characterizes individuals whose positive physical, cognitive, and socioemotional development is maintained longer, declining later in old age than is the case for most people.

For too long, only the declines that occur in late adulthood were highlighted, but recently there has been increased interest in the concept of successful aging (Perls, 2023). In a recent research review of adults 60 years of age and older, engaging in physical activity, having a positive outlook on life, being satisfied with their life, enjoying positive interpersonal relationships, participating in meaningful activities, and being financially secure emerged as the most important factors in their perception of experiencing successful aging (Bhattacharyya & others, 2023). In this review, the older adults did not consider living longer a primary component of successful aging.

Age and Happiness

Is there a best age to be? An increasing number of studies indicate that at least in the United States adults remain happy or become happier as they age. This has been called the "paradox of aging" because the physical health and function of older adults are often declining (Li, Lai, & Chappell, 2022). Consider also a U.S. study of approximately 28,000 individuals from 18 to 88 that revealed happiness increased with age (Yang, 2008). For example, about 33 percent described themselves as very happy at 88 years of age, compared with only about 24 percent in their late teens and early twenties. In a study of individuals from 22 to 93 years of age, older adults reported having more positive emotional experiences than did young adults (English & Carstensen, 2014).

Further, in a study of 18- to 76-year-olds conducted in April 2020, participants were asked about the frequency and intensity of their emotions over the past week, how risky they perceived COVID-19 to be to themselves and others, and how much time and opportunity they expected to have in the future (Carstensen, 2020). Older adults reported more frequent positive emotions and less intense negative emotions than younger adults, despite being more aware of the risks COVID-19 posed for them. The effect remained even though older adults estimated that they had less time left in life.

Why might more older people report being happier and more satisfied with their lives than younger people do? Despite the increase in physical problems and losses older adults experience, they are more contented with their lives, have better relationships with the people who matter to them, feel less pressure to achieve, have more time for leisurely pursuits, and have many years of experience that may help them adapt to their circumstances with greater wisdom than younger adults do (Carstensen & Hershfield, 2021).

Not all studies, though, have found an increase in life satisfaction with age (Steptoe, Deaton, & Stone, 2015). Some studies indicate that the lowest levels of life satisfaction are in middle age, especially from 45 to 54 years of age (OECD, 2014). Other studies

Below are five statements that you may agree or disagree with. Using the 1–7 scale below, indicate your agreement with each item by placing the appropriate number on the line preceding that item. Please be open and honest in your responding.

Response	Statement
Scale 7 Strongly agree 6 Agree 5 Slightly agree 4 Neither agree nor disagree 3 Slightly disagree 2 Disagree 1 Strongly disagree	_____ In most ways my life is close to my ideal.
	_____ The conditions of my life are excellent.
	_____ I am satisfied with my life.
	_____ So far I have gotten the important things I want in life.
	_____ If I could live my life over, I would change almost nothing.
	_____ Total score

Scoring
31–35 Extremely satisfied
26–30 Satisfied
21–25 Slightly satisfied
20 Neutral
15–19 Slightly dissatisfied
10–14 Dissatisfied
5–9 Extremely dissatisfied

Figure 4 How Satisfied Am I with My Life?

Source: E. Diener, R. A. Emmons, R. J. Larson, & S. Griffin. "The Satisfaction with Life Scale." *Journal of Personality Assessment*, 48, 1985, 71–75.

have found that life satisfaction varies across some countries. For example, research participants in the former Soviet Union and Eastern Europe, as well as those from South American countries, report a decrease in life satisfaction with advancing age (Deaton, 2008). Also, a recent study across 150 countries found that health was a better predictor of life satisfaction in individuals 58 years and older than in younger age groups (Joshani & Joyanovic, 2021).

Now that you have read about age variations in life satisfaction, think about how satisfied you are with your life. To help you answer this question, complete the items in Figure 4, which presents the most widely used measure in research on life satisfaction (Diener, 2023).

Developmental Issues

Was Ted Kaczynski born a killer, or did the events in his life turn him into one? Kaczynski himself

thought that his childhood was the root of his troubles. He said he grew up as a genius in a boy's body and never fit in with other children. Did his early experiences determine his later life? Is your own journey through life marked out ahead of time, or can your experiences change your path? Are the experiences you have early in your journey more important than later ones? Is your journey more like taking an elevator up a skyscraper with distinct stops along the way or more like a cruise down a river with smoother ebbs and flows? These questions point to three issues about the nature of development: the roles played by nature and nurture, stability and change, and continuity and discontinuity.

Nature and Nurture

The **nature-nurture issue** concerns the extent to which development is influenced by nature and by nurture. *Nature* refers to an organism's biological inheritance, *nurture* to its environmental experiences.

According to those who emphasize the role of nature, just as a sunflower grows in an orderly way—unless flattened by an unfriendly environment—so too a human grows in an orderly way. An evolutionary and genetic foundation produces commonalities in growth and development (Brooker & others, 2023). We walk before we talk, speak one word before two words, grow rapidly in infancy and less so in early childhood, experience a rush of sex hormones in puberty, reach the peak of our physical strength in late adolescence and early adulthood, and then physically decline. Proponents of the importance of nature acknowledge that extreme environments—those that are psychologically barren or hostile—can depress development. However, they believe that basic growth tendencies are genetically programmed into humans (Raven & others, 2023).

By contrast, other psychologists emphasize the importance of nurture, or environmental experiences, in development (Soria & Lawton, 2023).

nature–nurture issue The debate about the extent to which development is influenced by nature and by nurture. Nature refers to an organism's biological inheritance, nurture to its environmental experiences.

Experiences run the gamut from the individual's biological environment (nutrition, exercise, medical care, drugs, and physical accidents) to the social environment (family, peers, schools, community, media, and culture).

There has been a dramatic increase in the number of studies that reflect the *epigenetic view*, which states that development reflects an ongoing, bidirectional interchange between genes and the environment. These studies involve specific DNA sequences (Harvanek & others, 2023; Martin & others, 2022). The epigenetic mechanisms involve the actual molecular modification of the DNA strand as a result of environmental inputs in ways that alter gene functioning. In "Biological Beginnings" we will explore the epigenetic approach in greater depth.

Stability and Change

Is a shy child who hides behind the sofa when visitors arrive destined to become a wallflower at college dances, or might the child become a sociable, talkative individual? Is a fun-loving, carefree adolescent bound to have difficulty holding down a 9-to-5 job as an adult? These questions reflect the **stability-change issue**, involving the degree to which early traits and characteristics persist or change over time.

The roles of early and later experience are an aspect of the stability-change issue that has long been hotly debated (Allen, 2023). Some argue that warm, nurturant caregiving during infancy and toddlerhood predicts optimal development later in life (Waters, Waters, & Waters, 2021). The later-experience advocates see children as malleable throughout development and believe later sensitive caregiving is just as important as earlier sensitive caregiving.

Developmentalists who emphasize change take the more optimistic view that later experiences can produce change. Recall that in the life-span perspective, plasticity, the potential for change, exists throughout the life span. Experts such as Paul Baltes (2003) argue that older adults often show less capacity for learning new things than younger adults do. However, many older adults continue to be good at applying what they have learned in earlier times.



What are some key developmental issues?

Photodisc/Stockbyte/Getty Images

Continuity and Discontinuity

When developmental change occurs, is it gradual or abrupt? Think about your own development for a moment. Did you gradually become the person you are today? Or did you experience sudden, distinct changes in your growth? For the most part, developmentalists who emphasize nurture describe development as a gradual, continuous process. Those who emphasize nature often describe development as a series of distinct stages.

The **continuity-discontinuity issue** focuses on the degree to which development involves either gradual, cumulative change (continuity) or distinct stages (discontinuity). In terms of continuity, as the oak grows from a seedling to a giant tree, its development is continuous. Similarly, a child's first word, though seemingly an abrupt, discontinuous event, is actually the result of weeks and months of growth and practice. Puberty might seem abrupt, but it is a gradual process that occurs over several years.

In terms of discontinuity, as an insect grows from a caterpillar to a chrysalis to a butterfly, it passes through a sequence of stages in which change is qualitatively rather than quantitatively different. Similarly, at some point a child moves from not being able to think abstractly about the world to being able to do so. This is a qualitative, discontinuous change in development rather than a quantitative, continuous change.

stability-change issue The debate about the degree to which early traits and characteristics persist through life or change.

continuity-discontinuity issue The debate about the extent to which development involves gradual, cumulative change (continuity) or distinct stages (discontinuity).

Evaluating the Developmental Issues

Developmentalists generally acknowledge that development is not all nature or all nurture, not all stability or all change, and not all continuity or all discontinuity. Nature and nurture, stability and change, continuity and discontinuity characterize development throughout the life span.

Although most developmentalists do not take extreme positions on these three important issues, there is spirited debate regarding how strongly development is influenced by each of these factors (Ferrier, 2023).

Theories of Development

How can we answer questions about the roles of nature and nurture, stability and change, and continuity and discontinuity in development? How can we determine, for example, whether memory loss in older adults can be prevented or whether special care can repair the harm inflicted by child neglect? The scientific method is the best tool we have to answer such questions.

The scientific method is essentially a four-step process: (1) conceptualize a process or problem to be studied, (2) collect research information (data), (3) analyze data, and (4) draw conclusions.

In step 1, when researchers are formulating a problem to study, they often draw on theories and develop hypotheses. A **theory** is an interrelated, coherent set of ideas that helps to explain phenomena and make predictions. It may suggest **hypotheses**, which are specific assertions and predictions that can be tested. For example, a theory on mentoring might state that sustained support and guidance from an adult makes a difference in the lives of children from impoverished backgrounds because the mentor gives the children opportunities to observe and imitate the behavior and strategies of the mentor.

This section outlines five theoretical orientations to development: psychoanalytic, cognitive, behavioral and social cognitive, ethological, and ecological. These theories look at development from different perspectives, and they disagree about certain aspects of development. But many of their ideas are complementary, and each contributes an important piece to the life-span development puzzle. Although the theories disagree about certain aspects of development, together they let us see the total landscape of life-span development in all its richness.

Psychoanalytic Theories

Psychoanalytic theories describe development primarily in terms of unconscious (beyond awareness) processes that are heavily colored by emotion. Psychoanalytic theorists emphasize that behavior is merely a surface characteristic and that a true understanding of development requires analyzing the symbolic meanings of behavior and the deep inner workings of the mind. Psychoanalytic theorists also stress that early experiences with parents extensively shape development. These characteristics are highlighted in the main psychoanalytic theory, that of Sigmund Freud (1856–1939).

theory A coherent set of ideas that helps to explain data and to make predictions.

hypotheses Assertions or predictions, often derived from theories, that can be tested.

psychoanalytic theories Theories holding that development depends primarily on the unconscious mind and is heavily couched in emotion, that behavior is merely a surface characteristic, that it is important to analyze the symbolic meanings of behavior, and that early experiences are important in development.

Freud's Theory

Freud was a pioneer in the treatment of psychological problems. Based on his belief that patients who talked about their problems could be restored to psychological health, Freud developed a technique called psychoanalysis. As he listened to, probed, and analyzed his patients, he became convinced that their problems were the result of experiences early in life. He thought that as children grow up, their focus of pleasure and sexual impulses shifts from the mouth to the anus and eventually to the genitals. Consequently, he determined, we pass through five stages of

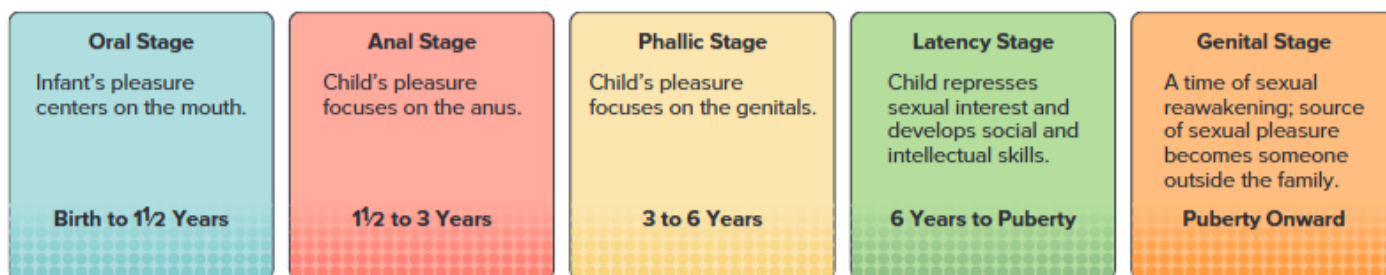


Figure 5 Freudian Stages.

Because Freud emphasized sexual motivation, his stages of development are known as psychosexual stages. In his view, if the need for pleasure at any stage is either undergratified or overgratified, an individual may become fixated, or locked in, at that stage of development.

psychosexual development: oral, anal, phallic, latency, and genital (see Figure 5). Our adult personality, Freud (1917) claimed, is determined by the way we resolve conflicts between sources of pleasure at each stage and the demands of reality.

Freud's followers significantly revised his psychoanalytic theory. Many of today's psychoanalytic theorists believe that Freud overemphasized sexual instincts; they place more emphasis on cultural experiences as determinants of an individual's development. Unconscious thought remains a central theme, but conscious thought plays a greater role than Freud envisioned. Next, we will outline the ideas of an important revisionist of Freud's theory—Erik Erikson.

Erikson's Psychosocial Theory

Erik Erikson recognized Freud's contributions but believed that Freud misjudged some important dimensions of human development. For one thing, Erikson (1950, 1968) said we develop in psychosocial stages, rather than the psychosexual stages that Freud described. According to Freud, the primary motivation for human behavior is sexual in nature; according to Erikson, motivation is social and reflects a desire to affiliate with other people. According to Freud, our basic personality is shaped in the first five years of life; according to Erikson, developmental change occurs throughout the life span. Thus, Freud viewed early experiences as far more important than later experiences, whereas Erikson emphasized the importance of both early and later experiences.

In Erikson's theory, eight stages of development unfold as we go through life (see Figure 6). At each stage, a unique developmental task confronts individuals with a crisis that must be resolved. According to Erikson, this crisis is not a catastrophe but a turning point marked by both increased vulnerability and enhanced potential. The more successfully an individual resolves these crises, the healthier his or her development will be.

Trust versus mistrust is Erikson's first psychosocial stage, which is experienced in the first year of life. Trust during infancy sets the stage for a lifelong expectation that the world will be a good and pleasant place to live.

Autonomy versus shame and doubt is Erikson's second stage. This stage occurs in late infancy and toddlerhood (1 to 3 years). After gaining trust in their caregivers, infants begin to discover that their behavior is their own. They start to assert their sense of independence or autonomy. They realize their will. If

Erikson's Stages	Developmental Period
Integrity versus despair	Late adulthood (60s onward)
Generativity versus stagnation	Middle adulthood (40s, 50s)
Intimacy versus isolation	Early adulthood (20s, 30s)
Identity versus identity confusion	Adolescence (10 to 20 years)
Industry versus inferiority	Middle and late childhood (elementary school years, 6 years to puberty)
Initiative versus guilt	Early childhood (preschool years, 3 to 5 years)
Autonomy versus shame and doubt	Infancy (1 to 3 years)
Trust versus mistrust	Infancy (first year)

Figure 6 Erikson's Eight Life-Span Stages.

Like Freud, Erikson proposed that individuals go through distinct, universal stages of development. In terms of the continuity-discontinuity issue, both favor the discontinuity side of the debate. Notice that the timing of Erikson's first four stages is similar to that of Freud's stages. *What are the implications of saying that people go through stages of development?*

Erikson's theory A psychoanalytic theory in which eight stages of psychosocial development unfold throughout the life span. Each stage consists of a unique developmental task that confronts individuals with a crisis that must be faced.



Erik Erikson with his wife, Joan, an artist. Erikson generated one of the most important developmental theories of the twentieth century.

Which stage of Erikson's theory are you in? Does Erikson's description of this stage characterize you?

Jon Erikson/Science Source

infants and toddlers are restrained too much or punished too harshly, they are likely to develop a sense of shame and doubt.

Initiative versus guilt, Erikson's third stage of development, occurs during the preschool years. As preschool children encounter a widening social world, they face new challenges that require active, purposeful, responsible behavior. Feelings of guilt may arise, though, if the child is irresponsible and is made to feel too anxious.

Industry versus inferiority is Erikson's fourth developmental stage, occurring approximately during the elementary school years. Children now need to direct their energy toward mastering knowledge and intellectual skills. The negative outcome is that the child may develop a sense of inferiority—feeling incompetent and unproductive.

During the adolescent years individuals face finding out who they are, what they are all about, and where they are going in life. This is Erikson's fifth developmental stage, *identity versus identity confusion*. If adolescents explore roles in a healthy manner and arrive at a positive path to follow in life, then they achieve a positive identity; if not, then identity confusion reigns.

Intimacy versus isolation is Erikson's sixth developmental stage, which individuals experience during early adulthood. At this time, individuals face the developmental task of forming intimate relationships. If young adults form healthy friendships and an intimate relationship with a partner, intimacy will be achieved; if not, isolation will result.

Generativity versus stagnation, Erikson's seventh developmental stage, occurs during middle adulthood. By generativity, Erikson means primarily a concern for helping the younger generation to develop and lead useful lives. The feeling of having done nothing to help the next generation is stagnation.

Integrity versus despair is Erikson's eighth and final stage of development, which individuals experience in late adulthood. During this stage, a person reflects on the past. If the person's life review reveals a life well spent, integrity will be achieved; if not, the retrospective glances likely will yield doubt or gloom—the despair Erikson described.

Evaluating Psychoanalytic Theories

Contributions of psychoanalytic theories like Freud's and Erikson's to life-span development include an emphasis on a developmental framework, family relationships, and unconscious aspects of the mind. These theories have been criticized for a lack of scientific support, too much emphasis on sexual underpinnings, and an image of people that is too negative.

Cognitive Theories

Whereas psychoanalytic theories stress the unconscious, cognitive theories emphasize conscious thoughts. Three important cognitive theories are Piaget's cognitive developmental theory, Vygotsky's sociocultural cognitive theory, and information-processing theory. All three focus on the development of complex thinking skills.

Piaget's Cognitive Developmental Theory

Piaget's theory states that children go through four stages of cognitive development as they actively construct their understanding of the world. Two processes underlie this cognitive construction of the world: organization and adaptation. To make sense of our world, we organize our experiences. For example, we separate important ideas from less important ideas, and we connect one idea to another. In addition to organizing our observations and experiences, we must adjust to changing environmental demands.

Piaget's theory The theory that children construct their understanding of the world and go through four stages of cognitive development.



Sensorimotor Stage

The infant constructs an understanding of the world by coordinating sensory experiences with physical actions. An infant progresses from reflexive, instinctual action at birth to the beginning of symbolic thought toward the end of the stage.

Birth to 2 Years of Age



Preoperational Stage

The child begins to represent the world with words and images. These words and images reflect increased symbolic thinking and go beyond the connection of sensory information and physical action.

2 to 7 Years of Age



Concrete Operational Stage

The child can now reason logically about concrete events and classify objects into different sets.

7 to 11 Years of Age



Formal Operational Stage

The adolescent reasons in more abstract, idealistic, and logical ways.

11 Years of Age Through Adulthood

Figure 7 Piaget's Four Stages of Cognitive Development.

According to Piaget, how a child thinks—not how much the child knows—determines the child's stage of cognitive development.

Left to right: Stockbyte/Getty Images; Jacobs Stock Photography/BananaStock/Getty Images; Fuse/Image100/Corbis; Hero Images Inc./Alamy Stock Photo



Jean Piaget, the famous Swiss developmental psychologist, changed the way we think about the development of children's minds. *What are some key ideas in Piaget's theory?*

Yves de Braine/Black Star/Stock Photo

Piaget (1954) described four stages in understanding the world (see Figure 7). Each stage is age-related and consists of a distinct way of thinking, a different way of understanding the world. Thus, according to Piaget, the child's cognition is *qualitatively* different in one stage compared with another. What are Piaget's four stages of cognitive development?

The *sensorimotor stage*, which lasts from birth to about 2 years of age, is the first Piagetian stage. In this stage, infants construct an understanding of the world by coordinating sensory experiences (such as seeing and hearing) with physical, motor actions—hence the term *sensorimotor*.

The *preoperational stage*, which lasts from approximately 2 to 7 years of age, is Piaget's second stage. In this stage, children begin to go beyond simply connecting sensory information with physical action and are now able to represent the world with words, images, and drawings. However, according to Piaget, preschool children still lack the ability to perform what he calls *operations*, which are internalized mental actions that allow children to do mentally what they previously could only do physically. For example, if you imagine putting two sticks together to see whether they would be as long as another stick, without actually moving the sticks, you are performing a concrete operation.

The *concrete operational stage*, which lasts from approximately 7 to 11 years of age, is the third Piagetian stage. In this stage, children can perform operations that involve objects, and they can reason logically about specific or concrete examples. Concrete operational thinkers, however, cannot imagine the steps necessary to complete an algebraic equation because doing so would require a level of thinking that is too abstract for this stage of development.

The *formal operational stage*, which appears between the ages of 11 and 15 and continues through adulthood, is Piaget's fourth and final stage. In this stage, individuals move beyond concrete experiences and think in abstract and more logical terms. As part of thinking more abstractly, adolescents develop images of ideal circumstances. They might think about what an ideal



Lev Vygotsky was born the same year as Piaget, but he died much earlier, at the age of 37. There is considerable interest today in Vygotsky's sociocultural cognitive theory of child development. *What are some key characteristics of Vygotsky's theory?*

A.R. Lauria/Dr. Michael Cole, Laboratory of Human Cognition, University of California, San Diego

parent is like and compare their parents to this ideal standard. They begin to entertain possibilities for the future and are fascinated with what they can become. In solving problems, they become more systematic, developing hypotheses about why something is happening the way it is and then testing these hypotheses. We will examine Piaget's cognitive developmental theory further in other chapters.

Vygotsky's Sociocultural Cognitive Theory

Like Piaget, the Russian developmentalist Lev Vygotsky (1896–1934) reasoned that children actively construct their knowledge. However, Vygotsky (1962) gave social interaction and culture far more important roles in cognitive development than Piaget did.

Vygotsky's theory is a sociocultural cognitive theory that emphasizes how culture and social interaction guide cognitive development. Vygotsky portrayed the child's development as inseparable from social and cultural activities (Valsiner, 2021). He stressed that cognitive development involves learning to use the inventions of society, such as language, mathematical systems, and memory strategies. Thus, in one culture children might learn to count with the help of a computer; in another they might learn by using beads. According to Vygotsky, children's social interaction with more-skilled adults and peers is indispensable to their cognitive development (Mecacci, 2021). Through this interaction, they learn to use the tools that will help them adapt and be successful in their culture. We will examine ideas about learning and teaching that are based on Vygotsky's theory in the chapter titled "Physical and Cognitive Development in Early Childhood."

Information-Processing Theory

Information-processing theory emphasizes that individuals manipulate information, monitor it, and strategize about it. Unlike Piaget's theory but like Vygotsky's theory, information-processing theory does not describe development as stage-like. Instead, according to this theory individuals develop a gradually increasing capacity for processing information, which allows them to acquire increasingly complex knowledge and skills (Hochmann & Kouider, 2022).

Robert Siegler (2006, 2017), a leading expert on children's information processing, states that thinking is information processing. In other words, when individuals perceive, encode, represent, store, and retrieve information, they are thinking. Siegler and his colleagues (Opfer & others, 2021; Siegler & Tian, 2022) emphasize that an important aspect of development is learning good strategies for processing information. For example, becoming a better reader might involve learning to monitor the key themes of the material being read.

Siegler (2006, 2017) also argues that the best way to understand how children learn is to observe them while they are learning. He emphasizes the importance of using the *microgenetic method* to obtain detailed information about processing mechanisms as they are occurring from moment to moment. Siegler concludes that most research methods indirectly assess cognitive change, being more like snapshots than movies. The microgenetic method seeks to discover not just what children know but also the cognitive processes involved in how they acquired the knowledge (Miller, 2015). A number of microgenetic studies have focused on a specific aspect of academic learning, such as how children learn whole number arithmetic, fractions, and other areas of math (Braithwaite & others, 2019).

The information processing approach often uses the computer as an analogy to help explain the connection between cognition and the brain

Vygotsky's theory A sociocultural cognitive theory that emphasizes how culture and social interaction guide cognitive development.

information-processing theory A theory emphasizing that individuals manipulate information, monitor it, and strategize about it. The processes of memory and thinking are central.

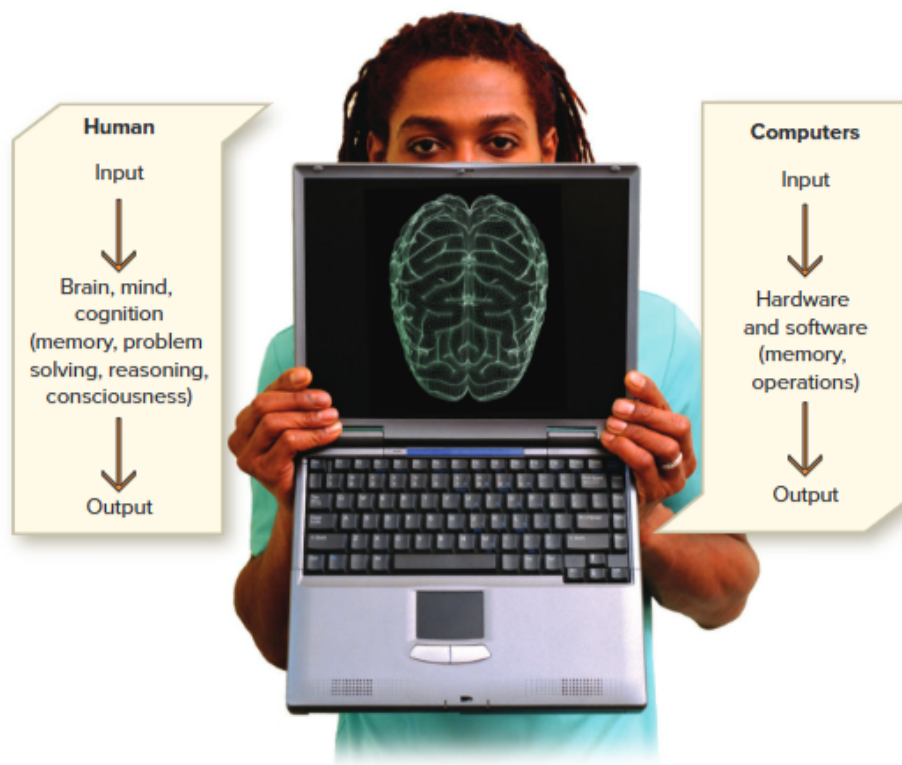


Figure 8 Comparing the Information Processing of Humans and Computers.

Psychologists who study cognition often use a computer analogy to explain how humans process information. The brain is analogous to the computer's hardware, and cognition is analogous to the computer's software.

Creatas/Getty Images

(Siegler & Alibali, 2020) (see Figure 8). The physical brain is described as the computer's hardware, and cognition as its software. In this analogy, the sensory and perceptual systems provide an "input channel," similar to the way data are entered into the computer. As input (information) comes into the mind, mental processes—or operations—act on it, just as the computer's software acts on the data. The transformed input generates information that remains in memory in much the same way that a computer stores what it has worked on. Finally, the information is retrieved from memory and "printed out" or "displayed" (so to speak) as an observable response.

Computers provide a logical and concrete, but oversimplified, model of the mind's processing of information. Inanimate computers and human brains function quite differently in some respects. For example, most computers receive information from a human who has already coded the information and removed much of its ambiguity. In contrast, each brain cell, or neuron, can respond to ambiguous information transmitted through sensory receptors such as the eyes and ears.

Computers can do some things better than humans. For instance, computers can perform complex numerical calculations much more rapidly and accurately than humans could. Computers can also apply and follow rules more consistently and with fewer errors than humans and can represent complex mathematical patterns better than humans.

Still, the brain's extraordinary capabilities will probably not be mimicked completely by computers at any time in the near future (Sternberg & Halpern, 2020). For example, although a computer can improve its ability to recognize patterns or use rules of thumb to make decisions, it does not have the means to develop new learning goals. Furthermore, the human mind is aware of itself; the computer is not. Indeed, no computer is likely to approach the richness of human consciousness.

Nonetheless, the computer's role in cognitive and developmental psychology continues to increase. An entire scientific field called *artificial intelligence (AI)* focuses on



The humanoid robot iCub was created by the Italian Institute of Technology to study such aspects of children's development as perception, cognition, and motor development. In this situation, the robot, the size of a 3.5-year-old child, is catching a ball. This robot is being used by more than 20 laboratories worldwide and has 53 motors that move the head, arms and hands, waist, and legs. It also can see and hear, as well as having the sense of proprioception (body configuration) and movement (using gyroscopes).

Marco Destefanis/Pacific Press/Sipa USA/Newscom

creating machines capable of performing activities that require intelligence when they are done by people (Aleem & others, 2023). And a new field titled *developmental robotics* uses robotic technology to explore various developmental topics and issues, such as motor development, perceptual development, information processing, and language development (Taniguchi & others, 2022). The hope is to build robots that are as much like humans as possible and in doing so to better understand how humans think and develop (Patacchiola & Cangelosi, 2022).

Evaluating Cognitive Theories

Contributions of cognitive theories include a positive view of development and an emphasis on the active construction of understanding. Criticisms include skepticism about the pureness of Piaget's stages and a belief that too little attention is paid to individual variations.

Behavioral and Social Cognitive Theories

Behavioral and social cognitive theories hold that development can be described in terms of behaviors learned through interactions with our surroundings. Behaviorism essentially holds that we can study scientifically only what can be directly observed and measured. Out of the behavioral tradition grew the belief that development is observable behavior that can be learned through experience with the environment (Miltner, 2023). In terms of the continuity-discontinuity issue discussed earlier in this chapter, the behavioral and social cognitive theories emphasize continuity in development and argue that development does not occur in stage-like fashion. Let's explore two versions of behaviorism: Skinner's operant conditioning and Bandura's social cognitive theory.

Skinner's Operant Conditioning

According to B. F. Skinner (1904–1990), through *operant conditioning* the consequences of a behavior produce changes in the probability of the behavior's recurrence. A behavior followed by a rewarding stimulus is more likely to recur, whereas a behavior followed by a punishing stimulus is less likely to recur. For example, when an adult smiles at a child after the child has done something, the child is more likely to engage in that behavior again than if the adult gives the child a disapproving look.

In Skinner's (1938) view, such rewards and punishments shape development. For Skinner the key aspect of development is behavior, not thoughts and feelings. He emphasized that development consists of the pattern of behavioral changes that are brought about by rewards and punishments. For example, Skinner would say that shy people learned to be shy as a result of experiences they had while growing up. It

follows that modifications to an environment can help a shy person become more socially oriented.

behavioral and social cognitive theories Theories holding that development can be described in terms of the behaviors learned through interactions with the environment.

social cognitive theory The theory that behavior, environment, and person/cognitive factors are important in understanding development.

Bandura's Social Cognitive Theory

Some psychologists agree with the behaviorists' notion that development is learned and is influenced strongly by environmental interactions. However, unlike Skinner, they also see cognition as important in understanding development. **Social cognitive theory** holds that behavior, environment, and person/cognitive factors are the key factors in development.

American psychologist Albert Bandura (1925–2021) was the leading architect of social cognitive theory. Bandura (1986, 2018) emphasized that cognitive processes have important links with the environment and behavior. His early research program focused heavily on *observational learning* (also called *imitation* or *modeling*), which is learning that occurs through observing what others do. For example, a young boy might observe his father yelling in anger and treating other people with hostility; and then later with his peers, the young boy acts very aggressively, showing the same behavioral characteristics as his father. Social cognitive theorists stress that people acquire a wide range of behaviors, thoughts, and feelings through observing others' behavior and that these observations form an important part of life-span development.

What is *cognitive* about observational learning, in Bandura's view? He proposes that people cognitively represent the behavior of others and then sometimes adopt this behavior themselves.

Bandura's (2018) most recent model of learning and development includes three elements: behavior, the person/cognition, and the environment.

An individual's confidence in being able to control his or her success is an example of a person factor; strategies for achieving success are an example of a cognitive factor. Bandura (2018) described *forethought* as a key cognitive factor in his social cognitive theory. When engag-

ing in *forethought*, individuals guide and motivate themselves by creating action plans, formulating goals, and visualizing positive outcomes of their actions. As shown in Figure 9, influences from behavior, person/cognition, and environment operate interactively.

Evaluating Behavioral and Social Cognitive Theories

Contributions of the behavioral and social cognitive theories include an emphasis on scientific research and environmental determinants of behavior. These theories have been criticized for placing too little emphasis on cognition (Skinner) and giving inadequate attention to developmental changes.

Ethological Theory

Ethology is the study of the behavior of animals in their natural habitat. Ethological theory stresses that behavior is strongly influenced by biology, is tied to evolution, and is characterized by critical or sensitive periods (Bateson, 2015). These are specific time frames during which, according to ethologists, the presence or absence of certain experiences has a long-lasting influence on individuals.

Lorenz's Research with Greylag Geese

European zoologist Konrad Lorenz (1903–1989) helped bring ethology to prominence. In his best-known research, Lorenz (1965) studied the behavior of greylag geese, which follow their mother as soon as they hatch. Lorenz separated the eggs laid by one goose into two groups. One group was returned to the goose to be hatched by her. The other group was



How Would You...?

As a **social worker**, how might you guide an unemployed adult client to use **forethought** to assist in finding employment?



Albert Bandura (1925–2021) was one of the leading architects of social cognitive theory. *How does Bandura's theory differ from Skinner's?*
Courtesy of Dr. Albert Bandura

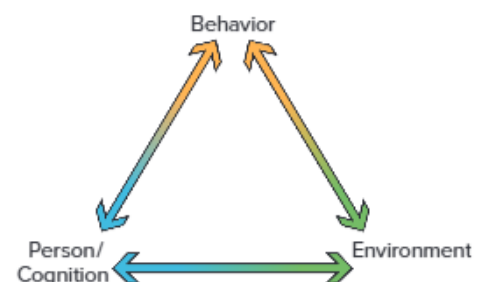


Figure 9 Bandura's Social Cognitive Model. The arrows illustrate how relations between behavior, person/cognition, and environment are reciprocal rather than one-way. Person/cognition refers to cognitive processes (for example, thinking and planning) and personal characteristics (for example, believing that you can control your experiences).

ethology An approach stressing that behavior is strongly influenced by biology, tied to evolution, and characterized by critical or sensitive periods.



Konrad Lorenz, a pioneering student of animal behavior, is followed through the water by three imprinted greylag geese.

Describe Lorenz's experiment with the geese. Do you think his experiment would have the same results with human babies? Explain.

Nina Leen/The LIFE Picture Collection/Shutterstock

hatched in an incubator. The goslings in the first group performed as predicted. They followed their mother as soon as they hatched. However, those in the second group, which saw Lorenz when they first hatched, followed him everywhere as though he were their mother. Lorenz marked the goslings and then placed both groups under a box. Mother goose and “mother” Lorenz stood aside as the box was lifted. Each group of goslings went directly to its “mother.” Lorenz called this

process *imprinting*—the rapid, innate learning that involves attachment to the first moving object seen.

John Bowlby (1969, 1989) illustrated an important application of ethological theory to human development. Bowlby stressed that attachment to a caregiver over the first year of life has important consequences throughout the life span. In his view, if this attachment is positive and secure, the individual will likely develop positively in childhood and adulthood. If the attachment is negative and insecure, development will likely not be optimal. Later we will explore the concept of infant attachment in much greater detail.

In Lorenz's view, imprinting needs to take place at a specific, very early time in the life of the animal, or else it will not take place. This point in time is called a critical period. A related concept is that of a sensitive period, and an example is the time during infancy when, according to Bowlby, attachment should occur in order to promote optimal development of social relationships.

Another theory that emphasizes biological foundations of development—evolutionary psychology—is presented in the chapter on “Biological Beginnings,” along with views on the role of heredity in development (Willey, Sandman, & Wood, 2023). In addition, we examine a number of biological theories of aging in the chapter on “Physical and Cognitive Development in Late Adulthood” (Sherazi & others, 2023).

Evaluating Ethological Theory

Contributions of ethological theory include a focus on the biological and evolutionary basis of development, and the use of careful observations in naturalistic settings. Criticisms include a belief that it places too much emphasis on biological foundations and that the concept of a critical and sensitive period might be too rigid.

Ecological Theory

While ethological theory stresses biological factors, ecological theory emphasizes environmental factors. One ecological theory that has important implications for understanding life-span development was created by Urie Bronfenbrenner (1917–2005).

Bronfenbrenner's ecological theory Bronfenbrenner's environmental systems theory, which focuses on five environmental systems: microsystem, mesosystem, exosystem, macrosystem, and chronosystem.

Bronfenbrenner's Ecological Theory

Bronfenbrenner's ecological theory (1986, 2004; Bronfenbrenner & Morris, 2006) holds that development reflects the influence of several environmental systems. The theory identifies five environmental systems: microsystem, mesosystem, exosystem, macrosystem, and chronosystem (see Figure 10).

The *microsystem* is the setting in which the individual lives. These contexts include the person's family, peers, school, and neighborhood. It is in the microsystem that the most direct interactions with social agents take place—with parents, peers, and teachers, for example. The individual is not a passive recipient of experiences in these settings, but someone who helps to construct the settings.

The *mesosystem* involves relations between microsystems or connections between contexts. Examples are the relation of family experiences to school experiences, school experiences to church experiences, and family experiences to peer experiences. For example, children whose parents have rejected them may have difficulty developing positive relations with teachers.

The *exosystem* consists of links between a social setting in which the individual does not have an active role and the individual's immediate context. For example, a husband's or child's experience at home may be influenced by a mother's experiences at work. The mother might receive a promotion that requires more travel, which might increase conflict with the husband and change patterns of interaction with the child.

The *macrosystem* involves the culture in which individuals live. Remember from earlier in the chapter that *culture* refers to the behavior patterns, beliefs, and all other products of a group of people that are passed from each generation to the next.

Remember also that cross-cultural studies—the comparison of one culture with one or more other cultures—provide information about the generality of development.

The *chronosystem* consists of the patterning of environmental events and transitions over the life course, as well as sociohistorical circumstances. For example, divorce is one transition. Researchers have found that the negative effects of divorce on children often peak in the first year after the divorce (Hetherington, 2006). By two years after the divorce, family interaction has become more stable. As an example of sociohistorical circumstances, consider how the opportunities for women to pursue a career have increased since the 1960s.

Responding to growing interest in biological contributions to development, Bronfenbrenner (2004) added biological influences to his theory and relabeled it as a bioecological theory. Nonetheless, it is still dominated by ecological, environmental contexts (Golinkoff & others, 2017).

Evaluating Ecological Theory

Contributions of ecological theory include its systematic examination of macro and micro dimensions of environmental systems

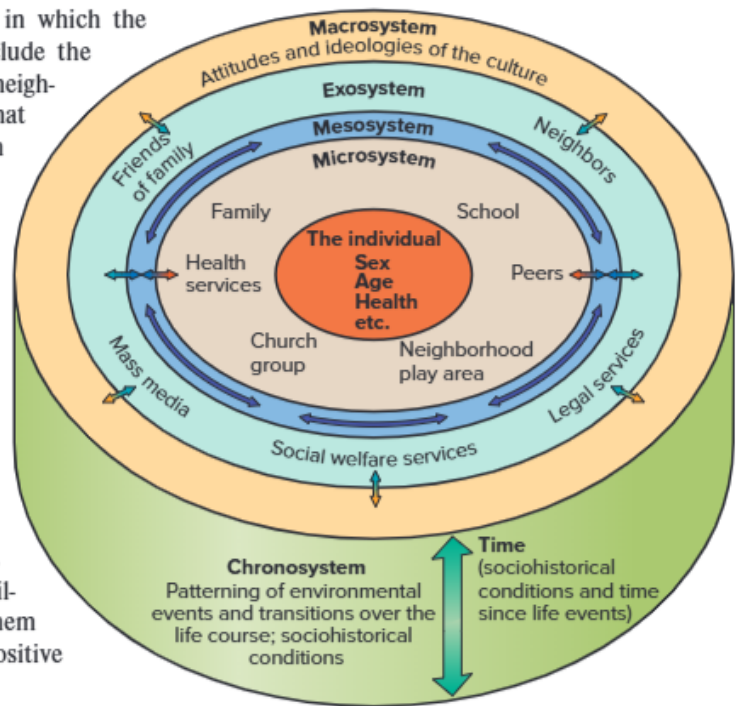
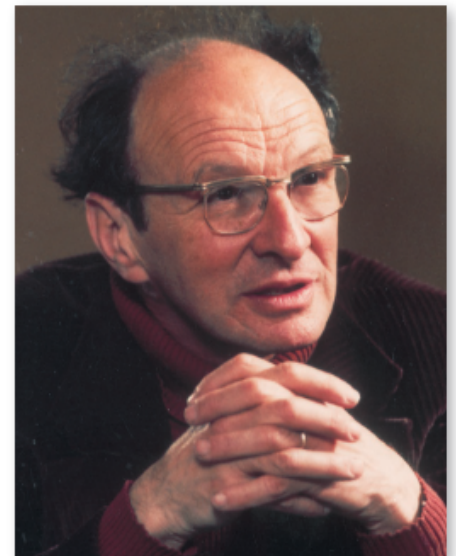


Figure 10 Bronfenbrenner's Ecological Theory of Development. Bronfenbrenner's ecological theory consists of five environmental systems: microsystem, mesosystem, exosystem, macrosystem, and chronosystem.



How Would You...?

As an educator, how might you use Bronfenbrenner's model to illustrate the need to look beyond the classroom to understand a student's poor performance?



Urie Bronfenbrenner developed ecological theory, a perspective that is receiving increased attention today. His theory emphasizes the importance of both micro and macro dimensions of the environment in which the child lives. Courtesy of Cornell University Photography

eclectic theoretical orientation
An approach that selects and uses whatever is considered the best in many theories.

and its attention to connections between environmental systems. A further contribution of Bronfenbrenner's theory is its emphasis on a range of social contexts beyond the family, such as peer relations, neighborhood, religious, school, workplace, and cultural environments, as factors influencing children's and adolescents' development (Matsumoto, Juang, & Hwang, 2023). For example, consider a recent analysis using a number of environmental systems in Bronfenbrenner's theory to better understand Black/White disparities in maternal and child health status (Noursi, Saluja, & Richey, 2021). In this analysis, root causes of these disparities were described as occurring at the microsystem (individual interaction), mesosystem (connecting environments), exosystem (community), and macrosystem (societal) levels of influence. Such circumstances can contribute to ethnic/racial disparities in preconception health and prenatal care, as well as health provider bias (among obstetricians, for example) and substandard quality of hospitals in predominantly Black localities.

Ecological theory has been criticized for giving inadequate attention to biological factors, as well as placing too little emphasis on cognitive factors.

An Eclectic Theoretical Orientation

No single theory described in this chapter can explain entirely the rich complexity of life-span development, but each has contributed to our understanding of development. Psychoanalytic theory highlights the importance of the unconscious mind. Erikson's theory best describes the changes that occur in adult development. Piaget's, Vygotsky's, and the information-processing views provide the most complete description of cognitive development. The behavioral and social cognitive and ecological theories have been the most adept at examining the environmental determinants of development. The ethological theories have drawn attention to biology's role and the importance of sensitive periods in development.

In short, although theories are helpful guides, relying on a single theory to explain development is probably a mistake. Instead, we will take an **eclectic theoretical orientation**, which does not follow any one theoretical approach but rather presents what are considered the best features of each theory. In this way, it represents the study of development as it actually exists—with different theorists making different assumptions, stressing different problems, and using different strategies to discover information. Figure 11 compares the main theoretical perspectives in terms of how they view important issues in life-span development.

THEORY	ISSUES	
	Continuity/discontinuity, early versus later experiences	Biological and environmental factors
Psychoanalytic	Discontinuity between stages—continuity between early experiences and later development; early experiences very important; later changes in development emphasized in Erikson's theory	Freud's biological determination interacting with early family experiences; Erikson's more balanced biological-cultural interaction perspective
Cognitive	Discontinuity between stages in Piaget's theory; continuity between early experiences and later development in Piaget's and Vygotsky's theories; no stages in Vygotsky's theory or information-processing theory	Piaget's emphasis on interaction and adaptation; environment provides the setting for cognitive structures to develop; information-processing view has not addressed this issue extensively but mainly emphasizes biological-environmental interaction
Behavioral and social cognitive	Continuity (no stages); experience at all points of development important	Environment viewed as the cause of behavior in both views
Ethological	Discontinuity but no stages; critical or sensitive periods emphasized; early experiences very important	Strong biological view
Ecological	Little attention to continuity/discontinuity; change emphasized more than stability	Strong environmental view

Figure 11 Summary of Theories and Issues in Life-Span Development.

Research in Life-Span Development

How do scholars and researchers with an eclectic orientation determine that one theory is somehow better than a different theory? The scientific method discussed earlier in this chapter provides a guide. Through scientific research, theories are tested and refined (Steiner & others, 2023).

Generally, research in life-span development is designed to test hypotheses, which may be derived from the theories just described (Leong & others, 2023). Through research, theories are modified to reflect new data, and occasionally new theories arise. How are data about life-span development collected? What types of research designs are used to study life-span development? And what are some ethical considerations in conducting research on life-span development?

Methods for Collecting Data

Whether we are interested in studying attachment in infants, the cognitive skills of children, or social relationships in older adults, we can choose from several ways of collecting data (Cooper & others, 2023). Here we outline the measures most often used, beginning with observation.

Observation

Scientific observation requires an important set of skills (Bakeman & Quera, 2023). For observations to be effective, they must be systematic. We need to have some idea of what we are looking for. We have to know whom we are observing, when and where we will observe, how the observations will be made, and how they will be recorded.

Where should we make our observations? We have two choices: the laboratory and the everyday world.

When we observe scientifically, we often need to control certain factors that determine behavior but are not the focus of our inquiry. For this reason, some research in life-span development is conducted in a **laboratory**, a controlled setting where many of the complex factors of the “real world” are absent. For example, suppose you want to observe how children react when they see other people behaving aggressively. If you observe children in their homes or schools, you have no control over how much aggression the children observe, what kind of aggression they see, which people they see acting aggressively, or how other people treat the children. In contrast, if you observe the children in a laboratory, you can control these and other factors and therefore have more confidence about how to interpret your observations.

Laboratory research does have some drawbacks, however, including the following concerns: (1) It is almost impossible to conduct research without the participants’ knowing they are being studied; (2) the laboratory setting is unnatural and therefore can cause the participants to behave unnaturally; (3) people who are willing to come to a university laboratory may not fairly represent groups from diverse cultural backgrounds; (4) people who are unfamiliar with university settings or with the idea of “helping science” may be intimidated by the laboratory setting.

Naturalistic observation provides insights that we sometimes cannot attain in the laboratory (Neuman, 2020). **Naturalistic observation** means observing behavior

in real-world settings without attempting to manipulate or control the situation. Life-span researchers conduct naturalistic observations at sporting events, child-care centers, work settings, malls, and other places people live in and frequent.

laboratory A controlled setting in which research can take place.

naturalistic observation Observation that occurs in a real-world setting without any attempt to manipulate the situation.



What are some important strategies in conducting observational research with children?
Marmaduke St. John / Alamy Stock Photo

Naturalistic observation was used in one study that focused on conversations in a children's science museum (Crowley & others, 2001). When visiting exhibits at the museum with their children, parents were more than three times as likely to engage boys than girls in explanatory talk. The gender difference occurred regardless of whether the father, the mother, or both parents were with the child, although the gender difference was greatest for fathers' science explanations to sons and daughters. This finding suggests a gender bias that encourages boys more than girls to be interested in science.

Survey and Interview

Sometimes the best and quickest way to get information about people is to ask them for it. One technique is to interview them directly. A related method is administering a survey (sometimes referred to as a questionnaire) consisting of a standard set of questions designed to obtain people's self-reported attitudes or beliefs about a particular topic. Surveys are especially useful when information from many people is needed. In a good survey, the questions are clear and unbiased, allowing respondents to answer unambiguously.

Surveys and interviews can be used to study topics ranging from religious beliefs to sexual habits to attitudes about gun control to beliefs about how to improve schools. Surveys and interviews may be conducted in person, over the telephone, by mail, and over the Internet.

One problem with surveys and interviews is the tendency of participants to answer questions in a way that they think is socially acceptable or desirable rather than to say what they truly think or feel (Madill, 2023). For example, on a survey or in an interview some individuals might say that they do not take drugs even though they do.

Standardized Test

A **standardized test** has uniform procedures for administration and scoring. Many standardized tests allow performance comparisons; they provide information about individual differences among people (Watson, 2023). One example is the Stanford-Binet intelligence test, which is discussed in detail later.

Your score on the Stanford-Binet test tells you how your performance compares with that of thousands of other people who have taken the test.

One criticism of standardized tests is that they assume a person's behavior is consistent and stable, yet personality and intelligence—two primary targets of standardized testing—can vary with the situation. For example, a person may perform poorly on a standardized intelligence test in an office setting but score much higher at home, where he or she is less anxious.

Case Study

A **case study** is an in-depth look at a single individual. Case studies are performed mainly by mental health professionals when, for either practical or ethical reasons, the unique aspects of an individual's life cannot be duplicated and tested in other individuals. A case study provides information about one person's experiences; it may focus on nearly any aspect of the subject's life that helps the researcher understand the person's mind, behavior, or other attributes. A researcher may gather information for a case study from interviews and medical records. In later chapters we discuss vivid case studies, such as that of Michael Rehbein, who had much of the left side of his brain removed at 7 years of age to end severe epileptic seizures.



Mahatma Gandhi was the spiritual leader of India in the mid-twentieth century. Erik Erikson conducted an extensive case study of Gandhi's life to determine what contributed to his identity development. *What are some limitations of the case study approach?*
Bettmann/Getty Images

standardized test A test that is given with uniform procedures for administration and scoring.

case study An in-depth examination of an individual.

A case study can provide a dramatic, in-depth portrayal of an individual's life, but we must be cautious when generalizing from this information. The subject of a case study is unique, with a genetic makeup and personal history that no one else shares. In addition, case studies involve judgments of unknown reliability. Researchers who conduct case studies rarely check to see whether other professionals agree with their observations or findings.

Physiological Measures

Researchers are increasingly using physiological measures when they study development at different points in the life span (Freberg, 2024; Tassinari & others, 2023). A physiological measure that is increasingly being used is neuroimaging, especially *functional magnetic resonance imaging* (fMRI), in which electromagnetic waves are used to construct images of a person's brain tissue and biochemical activity (de Bettencourt & others, 2023).

Heart rate has been used as an indicator of infants' and children's development of perception, attention, and memory (Cainelli & others, 2022). Further, heart rate has been used as an index of different aspects of emotional development, such as stress, anxiety, and depression (Henze & others, 2023).

Cortisol is a hormone produced by the adrenal gland that is linked to the body's stress level and has been measured in studies of temperament, emotional reactivity, mood, peer relations, and child psychopathology (Kelsall & others, 2023).

Sophisticated eye-tracking equipment is now being used to provide more detailed information about infants' perception (Liversedge & others, 2023), attention (Basel & others, 2022), face processing (Johnels & others, 2022), cognitive processing in bilingual children (Enke & others, 2022), and autism (Putnam & others, 2023).

Yet another dramatic change in physiological methods is the advancement in methods to assess the actual units of hereditary information—genes—in studies of biological influences on development (Aranha & Vega-Mercado, 2023). For example, recent advances in gene assessment have revealed several specific genes that are linked to childhood obesity (Song & others, 2022). Also, in the chapter on physical and cognitive development in late adulthood you will read about the role of the ApoE4 gene in Alzheimer disease (Pires & Rego, 2023).

Research Designs

In addition to choosing a method for collecting data, you also need to select a research design to study life-span development. There are three main types of research designs: descriptive, correlational, and experimental.

Descriptive Research

All of the data-collection methods that we have discussed can be used in **descriptive research**, which aims to observe and record behavior. For example, a researcher might observe the extent to which people are altruistic or aggressive toward each other. By itself, descriptive research cannot prove what causes some phenomenon, but it can reveal important information about people's behavior and provide a basis for more scientific studies (Riley & LaMarre, 2023).

Correlational Research

In contrast with descriptive research, correlational research goes beyond describing phenomena by providing information that helps to predict how people will behave. In **correlational research**, the goal is to describe the strength of the relationship between two or more events or characteristics. The more strongly the two events are correlated (or related or associated), the more effectively we can predict one event from the other (Peck & Case, 2024).

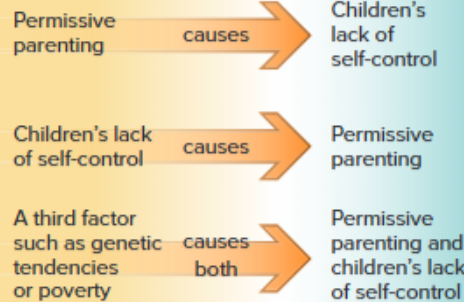
For example, to determine whether children of permissive parents have less self-control than other children, you would need to carefully record observations of parents' permissiveness and their children's self-control. You might observe that the higher a parent was in permissiveness,

descriptive research Type of research that aims to observe and record behavior.

correlational research A type of research that focuses on describing the strength of the relation between two or more events or characteristics.

Observed Correlation: As permissive parenting increases, children's self-control decreases.

Possible explanations for this observed correlation



An observed correlation between two events cannot be used to conclude that one event causes the second event. Other possibilities are that the second event causes the first event or that a third event causes the correlation between the first two events.

Figure 12 Possible Explanations for Correlational Data.

JupiterImages/Pixland/Getty Images

the lower the child was in self-control. You would then analyze these data statistically to yield a **correlation coefficient**, a number based on a statistical analysis that is used to describe the degree of association between two variables. Correlation coefficients range from -1.00 to $+1.00$. A negative number means an inverse relation. In the above example, you might find an inverse correlation between permissive parenting and children's self-control, with a coefficient of, say, $-.30$ meaning that parents who are permissive with their children are likely to have children who have low self-control. By contrast, you might find a positive correlation of $+.30$ between parental monitoring of children and children's self-control, meaning that parents who monitor their children effectively have children with good self-control.

The higher the correlation coefficient (whether positive or negative), the stronger the association between the two variables. A correlation of 0 means that there is no association between the variables. A correlation of $-.40$ is stronger than a correlation of $+.20$ because we disregard whether the correlation is positive or negative in determining the strength of the correlation.

A word of caution is in order, however. Correlation does not equal causation (Brase & Brase, 2024). The correlational finding just mentioned does not mean that permissive parenting necessarily causes low self-control in children. It could have that meaning, but it also could mean that a child's lack of self-control caused the parents to throw up their arms in despair and give up trying to control the child. It also could mean that other factors, such as heredity or poverty, caused the correlation between permissive parenting and low self-control in children. Figure 12 illustrates these possible interpretations of correlational data.

Experimental Research

To study causality, researchers turn to experimental research (Chambless & Hollon, 2023). An **experiment** is a carefully regulated procedure in which one or more factors believed to influence the behavior being studied are manipulated while all other factors are held constant. If the behavior under study changes when a factor is manipulated, we say that the manipulated factor has caused the behavior to change. In other words, the experiment has demonstrated cause and effect. The cause is the factor that was

manipulated. The effect is the behavior that changed because of the manipulation. Nonexperimental research methods (descriptive and correlational research) cannot establish cause and effect because they do not involve manipulating factors in a controlled way.

correlation coefficient A number based on statistical analysis that is used to describe the degree of association between two variables.

experiment A carefully regulated procedure in which one or more of the factors believed to influence the behavior being studied is manipulated and all other factors are held constant. Experimental research permits the determination of cause.

Independent and Dependent Variables Experiments include two types of changeable factors: independent and dependent variables. An *independent variable* is a manipulated, influential experimental factor. It is a potential cause. The label "independent" is used because this variable can be manipulated independently of other factors to determine its effect. An experiment may include one independent variable or several of them.

A *dependent variable* is a factor that can change in an experiment, in response to changes in the independent variable. As researchers manipulate the independent variable, they measure the dependent variable for any resulting effect.

For example, suppose that you wanted to study whether pregnant women could influence the breathing and sleeping patterns of their newborn babies by meditating during pregnancy. You might require one group of pregnant women to engage in a certain amount and type of meditation each day, while another group would not meditate; the meditation is thus the independent variable. When the infants are born, you would observe and measure their breathing and sleeping patterns. These patterns are the dependent variable, the factor that changes as the result of your manipulation.

Experimental and Control Groups Experiments can involve one or more experimental groups and one or more control groups. An experimental group is a group whose experience is manipulated. A control group is a comparison group that is as much like the experimental group as possible and that is treated in every way like the experimental group except for the manipulated factor (independent variable). The control group serves as a baseline against which the effects of the manipulated condition can be compared.

Random assignment is an important principle for deciding whether each participant will be placed in the experimental group or in the control group. Random assignment means that researchers assign participants to experimental and control groups by chance. It reduces the likelihood that the experiment's results will be due to any preexisting differences between groups. In the example of the effects of meditation by pregnant women on the breathing and sleeping patterns of their newborns, you would randomly assign half of the pregnant women to engage in meditation over a period of weeks (the experimental group) and the other half to not meditate over the same number of weeks (the control group). Figure 13 illustrates the nature of experimental research.

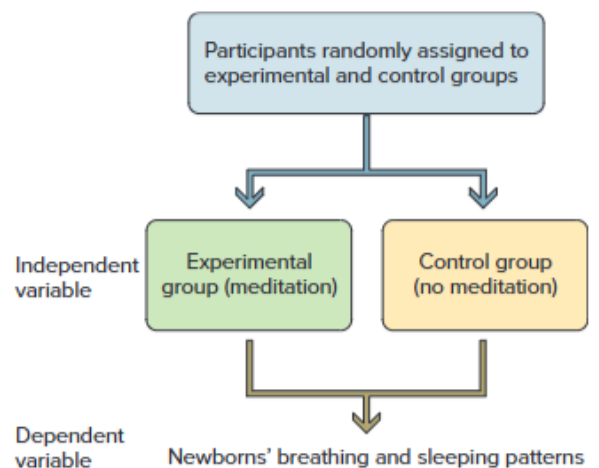


Figure 13 Principles of Experimental Research.

Imagine that you decide to conduct an experimental study of the effects of meditation by pregnant women on their newborns' breathing and sleeping patterns. You randomly assign pregnant women to experimental and control groups. The experimental-group women engage in meditation over a specified number of sessions and weeks. The control group does not. Then, when the infants are born, you assess their breathing and sleeping patterns. If the breathing and sleeping patterns of newborns whose mothers were in the experimental group are more positive than those of the control group, you conclude that meditation caused the positive effects.

Time Span of Research

Researchers in life-span development have a special concern with the relation between age and some other variable (Windle, 2023). To explore these relations, researchers can study different individuals of different ages and compare them, or they can study the same individuals as they age over time.

Cross-Sectional Approach

The **cross-sectional approach** is a research strategy that simultaneously compares individuals of different ages. A typical cross-sectional study might include three groups of children: 5-year-olds, 8-year-olds, and 11-year-olds. Another study might include groups of 15-year-olds, 25-year-olds, and 45-year-olds. The groups can be compared with respect to a variety of dependent variables, such as IQ, memory, peer relations, attachment to parents, hormonal changes, and so on. All of this can be accomplished in a short time. In some studies data are collected in a single day. Even in large-scale cross-sectional studies with hundreds of subjects, data collection does not usually take longer than several months to complete.

cross-sectional approach A research strategy in which individuals of different ages are compared at one time.

The main advantage of the cross-sectional study is that the researcher does not have to wait for the individuals to grow up or become older. Despite its efficiency, though, the cross-sectional approach has its drawbacks. It gives no information about how individuals change or about the stability of their characteristics. It can obscure the hills and valleys of growth and development. For example, a cross-sectional study of life satisfaction might reveal average increases and decreases, but it would not show how the life satisfaction of individual adults waxed and waned over the years. It also would not tell us whether the same adults who had positive or negative perceptions of life satisfaction in early adulthood maintained their relative degree of life satisfaction as they became middle-aged or older adults.

Longitudinal Approach

The **longitudinal approach** is a research strategy in which the same individuals are studied over a period of time, usually several years or more (Littlefield, 2023). For example, in a longitudinal study of life satisfaction, the same adults might be assessed periodically over a 70-year time span—at the ages of 20, 35, 45, 65, and 90, for example.

Longitudinal studies provide a wealth of information about vital issues such as stability and change in development and the importance of early experience for later development, but they do have drawbacks (Johal & others, 2023). They are expensive and time-consuming. The longer the study lasts, the more participants drop out—they move, get sick, lose interest, and so forth. The participants who remain may be dissimilar to those who drop out, biasing the outcome of the study. Those individuals who remain in a longitudinal study over a number of years may be more responsible and conformity-oriented than the ones who dropped out, for example, or they might lead more stable lives.

Cohort Effects

A *cohort* is a group of people who are born at a similar point in history and share similar experiences as a result, such as living through the Vietnam War or growing up in the same city around the same time. These shared experiences may produce a range of differences among cohorts (Schaie, 2016). For example, people who were teenagers during the Great Depression are likely to differ from people who were teenagers during the booming 1990s in their educational opportunities and economic status, in how they were raised, and in their attitudes toward sex and religion. In life-span development research, **cohort effects** are due to a person's time of birth, era, or generation but not to actual age (Yuen & others, 2023).

Cohort effects are important because they can powerfully affect the dependent measures in a study ostensibly concerned with age. Researchers have shown it is especially important to be aware of cohort effects when assessing adult intelligence (Schaie, 2016). Individuals born at different points in time—such as 1920, 1940, and 1960—have had varying opportunities for education. Individuals born in earlier years had less access to education, and this fact may have a significant effect on how this cohort performs on intelligence tests. Some researchers have found that cross-sectional studies indicate that more than 90 percent of cognitive decline in aging is due to a slowing of processing speed, whereas longitudinal studies reveal that 20 percent or less of cognitive decline is due to processing speed (MacDonald & others, 2003; MacDonald & Stawski, 2016).

Cross-sectional studies can show how different cohorts respond, but they can confuse age changes and cohort effects. Longitudinal studies are effective in studying age changes, but only within one cohort.

Various generations have been given labels by the popular culture. Figure 14 describes the labels of various generations, the historical period for each one, and the reasons for their labels.

Until recently, the latest generation was labeled millennials and described as being more technologically sophisticated and more ethnically diverse than earlier generations. However, the Pew Research Center,

longitudinal approach A research strategy in which the same individuals are studied over a period of time, usually several years or more.

cohort effects Effects that are due to a subject's time of birth or generation but not age.

Generation	Historical Period	Reasons for Label
Generation Z/ Post-Millennials	Individuals born in 1997 and later	More immersed in a technological world, more ethnically diverse, and better educated than millennials.
Millennials	Individuals born between 1981 and 1996	First generation to come of age and enter emerging adulthood (18 to 25 years of age) in the twenty-first century (the new millennium). Two main characteristics: (1) connection to technology, and (2) ethnic diversity.
Generation X	Individuals born between 1965 and 1980	Described as lacking an identity and savvy loners.
Baby Boomers	Individuals born between 1946 and 1964	Label used because this generation represents the spike in the number of babies born after World War II; the largest generation ever to enter late adulthood in the United States.
Silent Generation	Individuals born between 1928 and 1945	Children of the Great Depression and World War II; described as conformists and civic minded.

Figure 14 Generations, Their Historical Periods, and Characteristics.

which has periodically assessed generational trends in recent decades, recently began to describe millennials as anyone born between 1981 and 1996, and anyone born in 1997 or later as part of a new generation (Dimock, 2019). Currently, there is no consensus on what the name of this new generation should be, but the main candidate is Generation Z. The oldest members of the new generation turned 23 in 2020 while the oldest millennials turned 40 in 2020. What characterizes this new generation? They are even more technologically sophisticated and ethnically diverse than millennials. Members of the new generation have grown up with always available/always on technological devices, are immersed in social media, and communicate with others online and through mobile devices far more than in person. Also, Generation Z individuals are even better educated than millennials, with more going to college and more living with a college-educated parent (Fry & Parker, 2018).

How Would You...?

As an **educator**, how would you explain the generational differences in viewpoints to a student, her mother, and her grandmother?



Conducting Ethical Research

Researchers who study human development and behavior confront many ethical issues (Crocker, 2023). For example, a developmentalist who wanted to study aggression in children would need to design a study in which no child would be harmed physically or psychologically, and the researcher would also need to get permission from the university to carry out the study. Then the researcher would have to explain the study to each child's parents and obtain consent for the child to participate. Ethics in research may affect you personally if you ever serve as a participant in a study. In that event, you need to know your rights as a participant and the responsibilities of researchers to ensure that these rights are safeguarded.

Today, proposed research at colleges and universities must pass the scrutiny of a research ethics committee before the research can begin. In addition, the American Psychological Association (APA) has developed a code of ethics that instructs psychologists to protect their research participants from mental and physical harm. The participants' best interests need to be kept foremost in the researcher's mind (Fried, 2023).

How does the youth experienced by Generation Z differ from that of earlier generations?

Hero Images/Alamy Stock Photo

APA's guidelines address four important issues:

1. **Informed consent**—All participants must know what their research participation will involve and what risks might develop. Even after informed consent is given, participants must retain the right to withdraw from the study at any time and for any reason.
2. **Confidentiality**—Researchers are responsible for keeping all of the data they gather on individuals completely confidential and, when possible, completely anonymous.
3. **Debriefing**—After the study has been completed, participants should be informed of its purpose and the methods that were used. In most cases, the experimenter also can inform participants in a general manner beforehand about the purpose of the research without leading participants to behave in a way they think that the experimenter is expecting.
4. **Deception**—In some circumstances, telling the participants beforehand what the research study is about substantially alters the participants' behavior and invalidates the researcher's data. In all cases of deception, however, the psychologist must ensure that the deception will not harm the participants and that the participants will be *debriefed* (told the complete nature of the study) as soon as possible after the study is completed.

Cultural and Ethnic Bias

Today there is a growing realization that research on life-span development needs to include more people from diverse ethnic groups (Clauss-Ehlers & others, 2023). Historically, people from ethnic minority groups (Black Americans, Latinos, Asian

Americans, and Native Americans) were excluded from most research in the United States and simply thought of as variations from the norm or average. If minority individuals were included in samples and their scores didn't fit the norm, they were viewed as confounds or "noise" in data and discounted. Given the fact that individuals from diverse ethnic groups were excluded from research on life-span development for so long, we might reasonably conclude that people's real lives are perhaps more varied than research data have indicated in the past.

Researchers also have tended to overgeneralize about ethnic groups (Halder & Squires, 2023). **Ethnic gloss** involves using an ethnic label in a superficial way that portrays an ethnic group as being more homogeneous than it really is (Trimble, 2022). For example, a researcher might describe a research sample like this: "The participants were 60 Latinos." A more complete description of the group might be something like this: "The 60 Latino participants were Mexican Americans from low-income neighborhoods in the southwestern area of Los Angeles. Thirty-six were from homes in which Spanish is the dominant language spoken, 24 from homes in which English is the main language spoken. Thirty were born in the United States, 30 in Mexico. Twenty-eight described themselves as Mexican American, 14 as Mexican, 12 as Latino, and 6 as American." Ethnic gloss can cause researchers to obtain samples of ethnic groups that are not representative of the group's diversity, which can lead to overgeneralization and stereotyping.

ethnic gloss Involves using an ethnic label in a superficial way that portrays an ethnic group as being more homogeneous than it really is.



Look at these two photographs, one of all non-Latino White males, the other of a diverse group of females and males from different ethnic groups, including some non-Latino White males. Consider a topic in life-span development, such as parenting, love, or cultural values. *If you were conducting research on this topic, might the results of the study differ depending on whether the participants in your study were the individuals in the photograph on the top or on the bottom?* (Top) Wayne Hutchinson/ Alamy Stock Photo; (Bottom) Punchstock/Digital Vision

In sum, studies of life-span development are most useful when they are conducted without bias or prejudice toward any particular group of people (Matsumoto & van de Vijver, 2023; Roosa & others, 2023).

Summary

The Life-Span Perspective

- Development is the pattern of change that begins at conception and continues through the life span. It includes both growth and decline.
- The life-span perspective includes these basic ideas: Development is lifelong, multidimensional, multidirectional, and plastic; its study is multidisciplinary; it is embedded in contexts; it involves growth, maintenance, and regulation; and it is a co-construction of biological, sociocultural, and individual factors.
- Health and well-being, parenting, education, sociocultural contexts and diversity, and social policy are all areas of contemporary concern for those who study life-span development.

The Nature of Development

- Three key developmental processes are biological, cognitive, and socioemotional. Development is influenced by an interplay of these processes.
- The life span is commonly divided into the prenatal period, infancy, early childhood, middle and late childhood, adolescence, early adulthood, middle adulthood, and late adulthood.
- We often think of age only in chronological terms, but a full evaluation of age also requires the consideration of biological age, psychological age, and social age.
- Three pathways of aging are pathological aging, normal aging, and successful aging.
- In research covering adolescence through late adulthood, many but not all studies find that older adults report the highest level of life satisfaction.
- Three important issues in the study of development are the nature-nurture issue, the continuity-discontinuity issue, and the stability-change issue.

Theories of Development

- According to psychoanalytic theories, including those of Freud and Erikson, development primarily depends on the unconscious mind and is heavily couched in emotion.
- Cognitive theories emphasize thinking, reasoning, language, and other cognitive processes. Three main cognitive theories are Piaget's, Vygotsky's, and information processing.
- Behavioral and social cognitive theories emphasize the environment's role in development. Two key behavioral and social cognitive theories are Skinner's operant conditioning and Bandura's social cognitive theory.
- Lorenz's ethological theory stresses the biological and evolutionary bases of development.
- According to Bronfenbrenner's ecological theory, development predominantly reflects the influence of five environmental systems—the microsystem, mesosystem, exosystem, macrosystem, and chronosystem.
- An eclectic orientation incorporates the best features of different theoretical approaches.

Research In Life-Span Development

- The main methods for collecting data about life-span development are observation, survey (questionnaire) or interview, standardized test, case study, and physiological measures.
- Three basic research designs are descriptive, correlational, and experimental.
- To examine the effects of time and age, researchers can conduct cross-sectional or longitudinal studies. Life-span researchers are especially concerned about cohort effects.
- Researchers have an ethical responsibility to safeguard the well-being of research participants.
- Researchers need to guard against cultural and ethnic bias in research.

Key Terms

behavioral and social cognitive theories	cross-cultural studies	hypotheses	Piaget's theory
biological processes	cross-sectional approach	information-processing theory	psychoanalytic theories
Bronfenbrenner's ecological theory	culture	laboratory	social cognitive theory
case study	descriptive research	life-span perspective	social policy
cognitive processes	development	longitudinal approach	socioeconomic status (SES)
cohort effects	eclectic theoretical orientation	naturalistic observation	socioemotional processes
context	Ekerson's theory	nature-nurture issue	stability-change issue
continuity-discontinuity issue	ethnic gloss	nonnormative life events	standardized test
correlation coefficient	ethnicity	normative age-graded influences	theory
correlational research	ethology	normative history-graded influences	Vygotsky's theory
	experiment		
	gender		